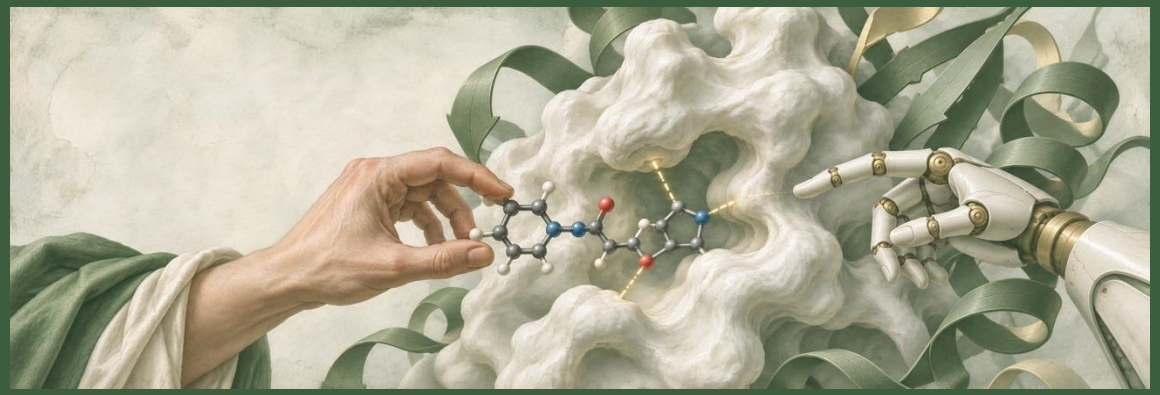
Workshop: Human vs AI

Ligand design and docking towards selectivity



Icaro Simon, Jody Pacalon & Kasper Harpsøe

Copenhagen, 10 November 2026

UNIVERSITY OF COPENHAGEN

SDU

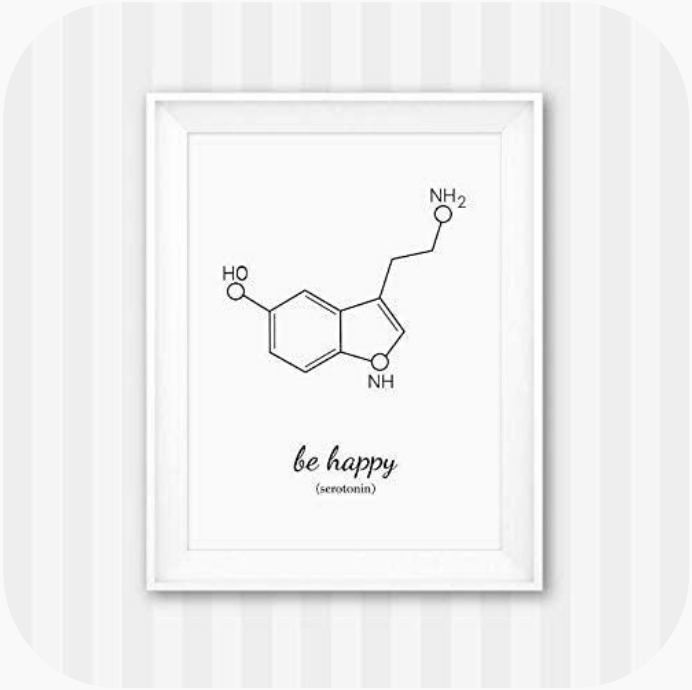
Center for Pharmaceutical Data Science Education

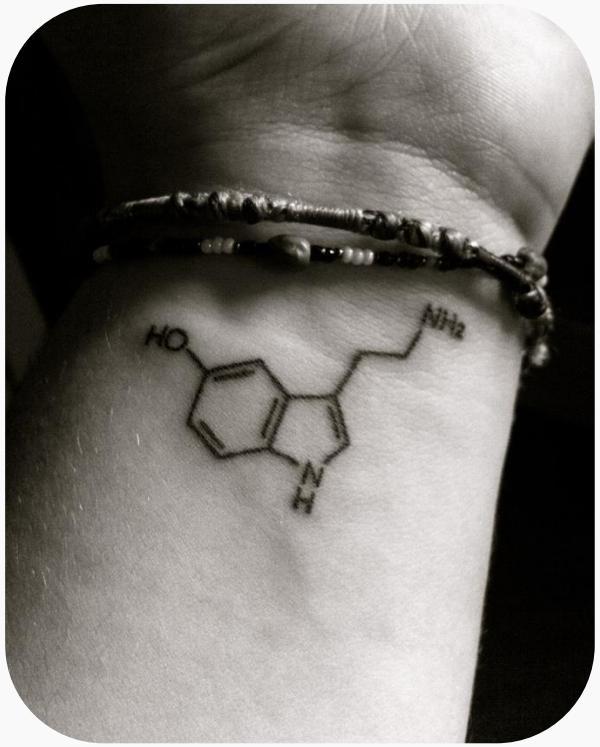
What you will learn in this workshop

- Diagnose and prepare experimental receptor structures.
- Align related receptors and compare ligand contacts and pocket shape.
- Turn structural observations into affinity and selectivity hypotheses.
- Draw molecules, check stereochemistry, and evaluate AI-generated SMILES.
- Prepare ligands, generate receptor grids, and dock to two targets.
- Interpret scores alongside poses and distinguish docking rankings from measured affinity.

The biological problem:
designing selective drugs

Serotonin and its receptors

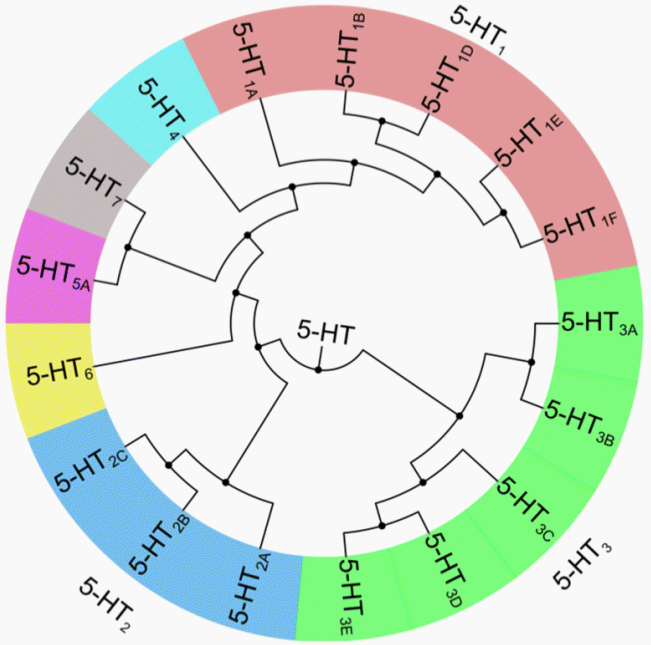




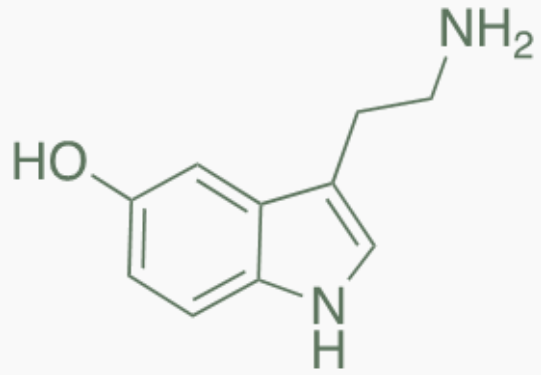
Serotonin or 5-hydroxytryptamine (5-HT)

14

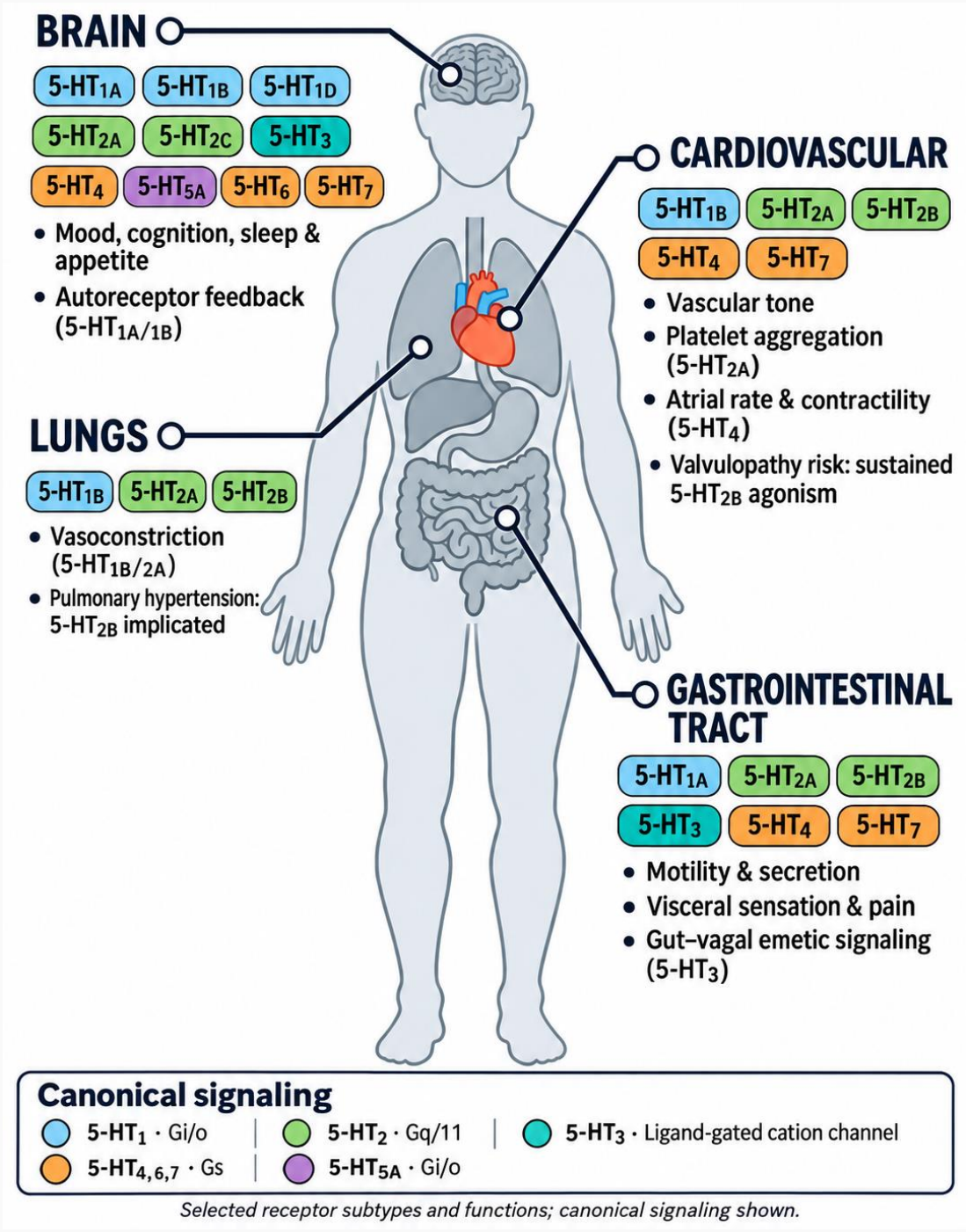
receptors



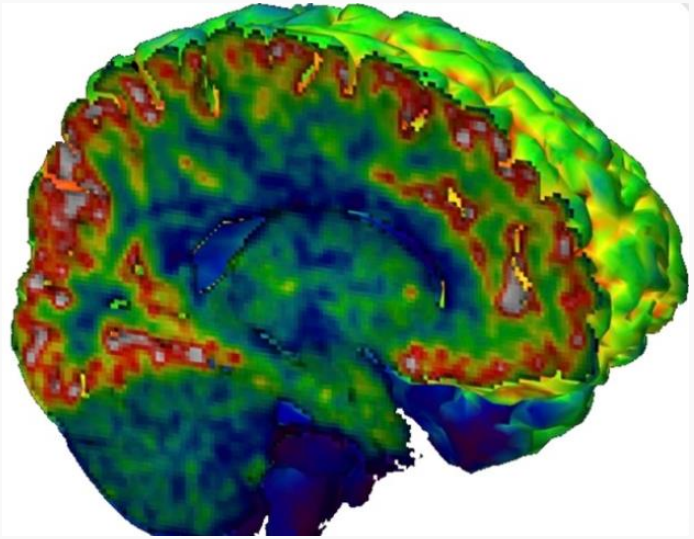
Many physiological roles



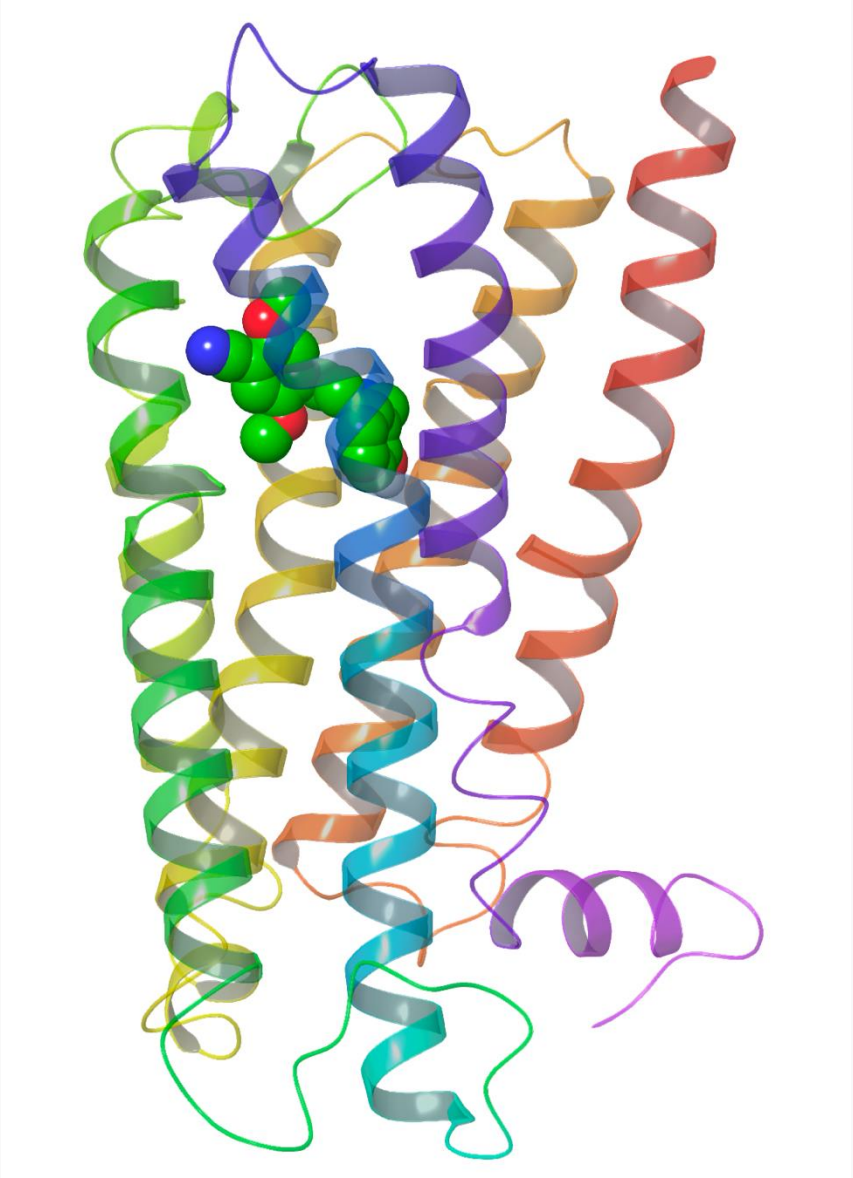
Serotonin
(5-hydroxytryptamine, 5-HT)
The oldest neurotransmitter



The 5-HT_{2A} Receptor



Distribution of 5-HT_{2A} in the human brain [1]



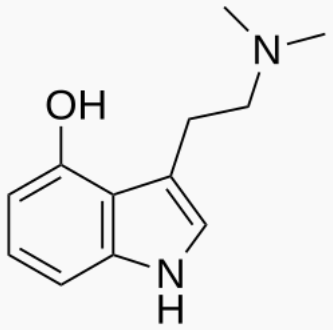
Structure of the $5HT_{2A}$ receptor bound to psychedelic 25CN-NBOH (PDB 6WHA)

Psychedelics

The oldest psychopharmacological agents (~3700 BCE)



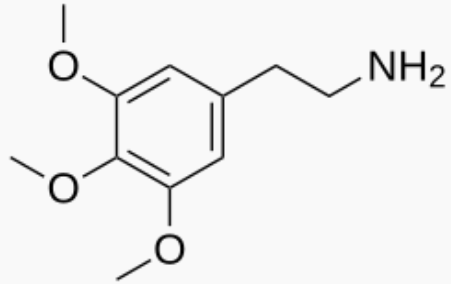
Psilocybin mushrooms
(Psilocybe cubensis)



Psilocin



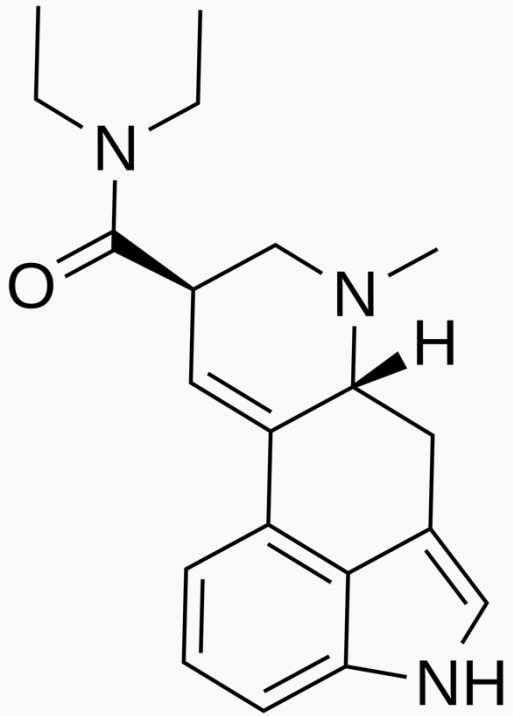
Peyote cactus
(Lophophora williamsii)



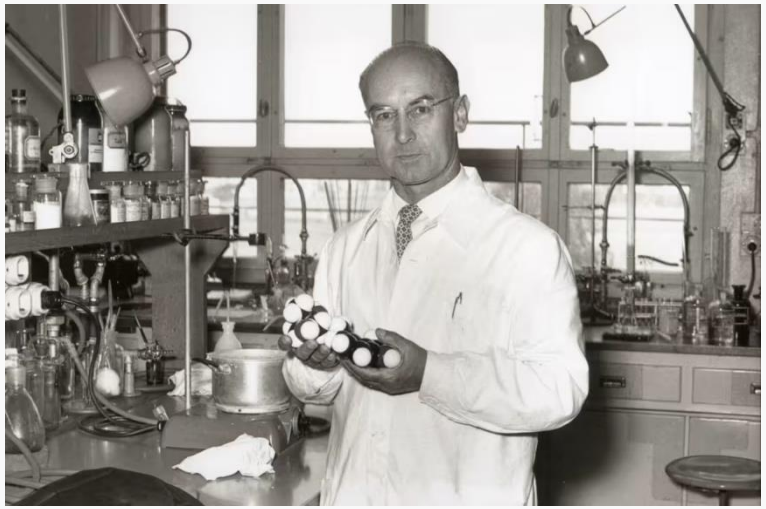
Mescaline

Photographs by Hans B. and Erik Fenderson, distributed under a CC BY-SA 3.0 license.

Lysergic acid diethylamine (LSD)



Structure of LSD



Swiss chemist Albert Hofmann (1906–2008)



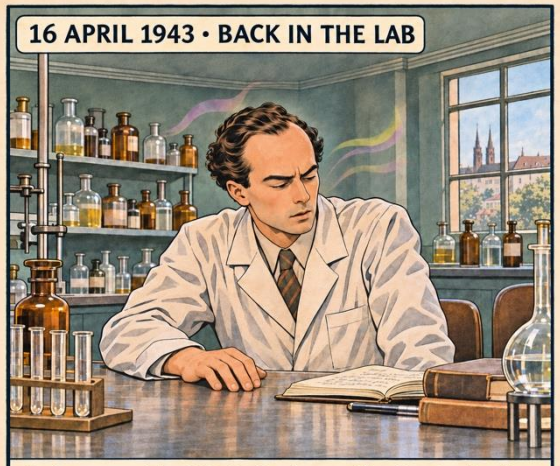
Albert Hofmann at Sandoz Laboratories, Basel (1938).

Source: House of Switzerland (https://houseofswitzerland.org/swiss-stories-archives/swiss-beginnings-lsd)

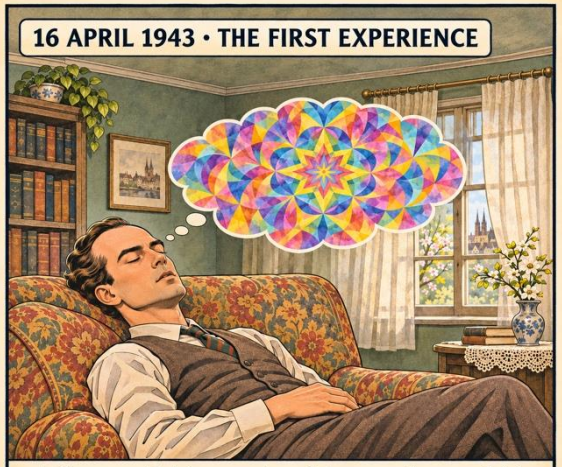
The discovery of LSD



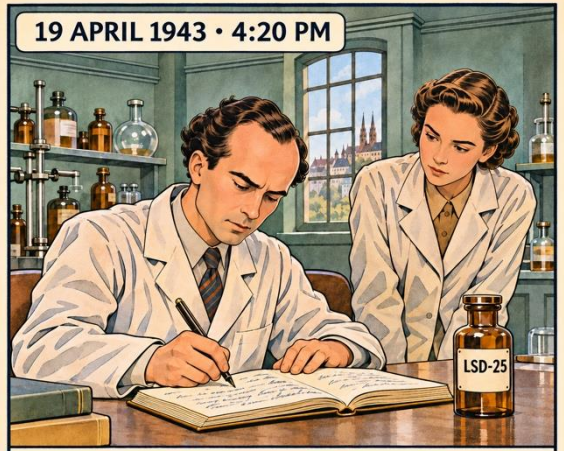
Seeking a circulatory and respiratory stimulant, Hofmann synthesizes LSD-25, his 25th lysergic acid derivative. Initial tests fail to justify further development, and the compound is shelved.



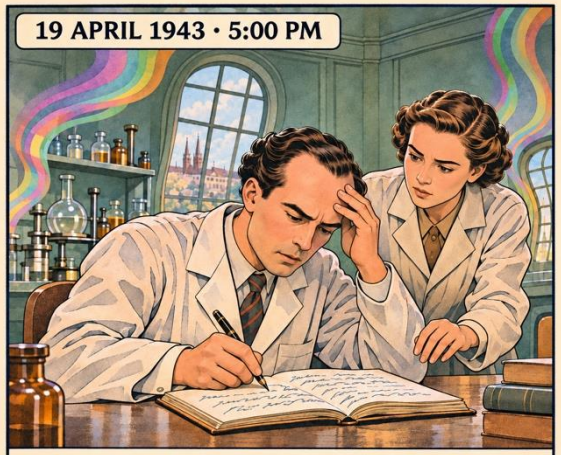
Hofmann revisits LSD-25. During purification, he develops restlessness and dizziness and leaves work early. He later suspects accidental exposure; the route remains uncertain.



At home, with his eyes closed, he experiences vivid, kaleidoscopic imagery. It fades after about two hours. He suspects LSD-25 caused the unexpected effects.



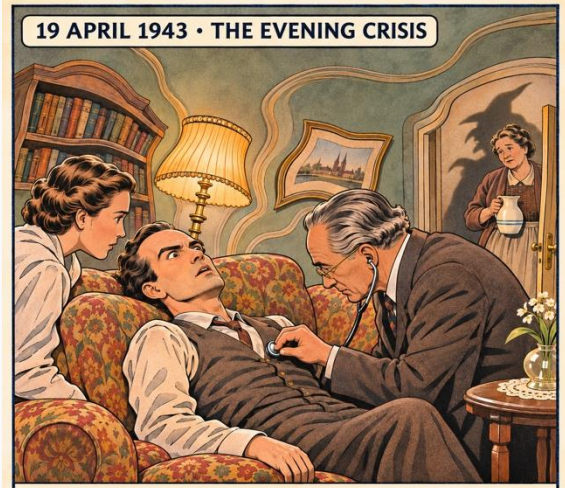
To test his suspicion, Hofmann deliberately takes 0.25 mg of LSD tartrate. He believes this a cautious starting amount, unaware of its extraordinary potency.



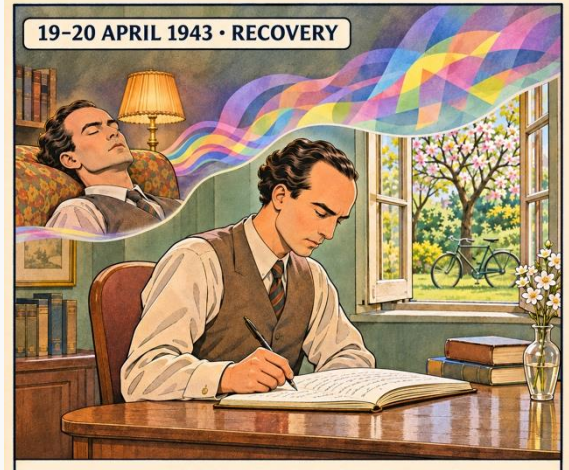
His notebook records dizziness, anxiety, visual disturbances and an urge to laugh. As writing becomes difficult, he asks Susi Ramstein to accompany him home.



With no car available because of wartime restrictions, they cycle home. The street seems distorted, and Hofmann feels almost stationary, although they are moving quickly.



At home, he fears poisoning or madness and perceives a neighbor as a witch. A doctor examines him; the terrifying phase gradually eases.



Later, sounds evoke shifting colors as the fear subsides. After sleep, he wakes refreshed on April 20 and documents the experience that revealed LSD's remarkable psychoactive effects.

A new therapeutic frontier

Article

Efficacy and safety of low- versus high-dose-LSD-assisted therapy in patients with major depression: A randomized trial

Felix Müller,[1,2,5,*] Hannes Zaczek,[1] Anna M. Becker,[2] Laura Ley,[2] Stefan Borgwardt,[3] Joyce Santos de Jesus,[2] Nico Loh,[1] Jan Kohut,[1] Mathias Auernig,[1] Christopher Boehlke,[1,4] and Matthias E. Liechti[2]
[1]University of Basel, Department of Psychiatry (UPK), 4012 Basel, Switzerland
[2]University Hospital Basel and University of Basel, Division of Clinical Pharmacology and Toxicology, Department of Biomedicine and Department of Clinical Research, 4031 Basel, Switzerland
[3]Department of Psychiatry and Psychotherapy, University of Lübeck, 23538 Lübeck, Germany
[4]University Hospital Basel, Department of Palliative Care, 4031 Basel, Switzerland
[5]Lead contact
*Correspondence: felix.mueller@upk.ch
https://doi.org/10.1016/j.medj.2025.100725

Priority Communication

Biological Psychiatry

Lysergic Acid Diethylamide–Assisted Therapy in Patients With Anxiety With and Without a Life-Threatening Illness: A Randomized, Double-Blind, Placebo-Controlled Phase II Study

Friederike Holze, Peter Gasser, Felix Müller, Patrick C. Dolder, and Matthias E. Liechti

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 NOVEMBER 3, 2022 VOL. 387 NO. 18

Single-Dose Psilocybin for a Treatment-Resistant Episode of Major Depression

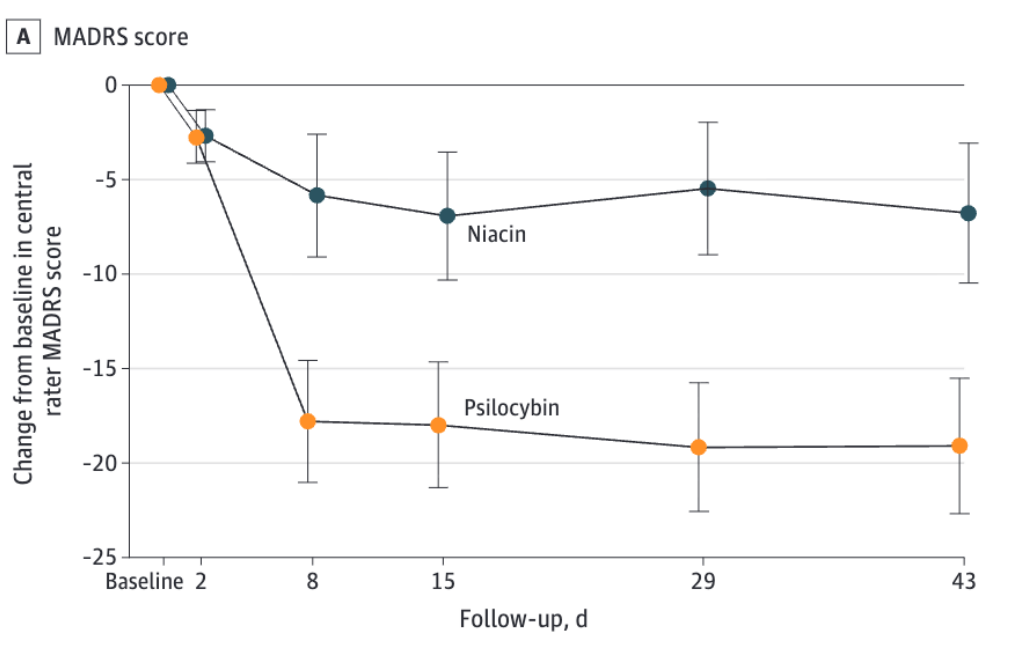
G.M. Goodwin, S.T. Aaronson, O. Alvarez, P.C. Arden, A. Baker, J.C. Bennett, C. Bird, R.E. Blom, C. Brennan, D. Brusch, L. Burke, K. Campbell-Coker, R. Carhart-Harris, J. Cattell, A. Daniel, C. DeBattista, B.W. Dunlop, K. Eisen, D. Feifel, M.K. Forbes, H.M. Haumann, D.J. Hellerstein, A.I. Hoppe, M.I. Husain, L.A. Jelen, J. Kamphuis, J. Kawasaki, J.R. Kelly, R.E. Key, R. Kishon, S. Knatz Peck, G. Knight, M.H.B. Koolen, M. Lean, R.W. Licht, J.L. Maples-Keller, J. Mars, L. Marwood, M.C. McElhiney, T.L. Miller, A. Mirow, S. Mistry, T. Mletzko-Crowe, L.N. Modlin, R.E. Nielsen, E.M. Nielson, S.R. Offerhaus, V. O'Keane, T. Páleníček, D. Printz, M.C. Rademaker, A. van Reemst, F. Reinholdt, D. Repantis, J. Rucker, S. Rudow, S. Ruffell, A.J. Rush, R.A. Schoevers, M. Seynaeve, S. Shao, J.C. Soares, M. Somers, S.C. Stansfield, D. Sterling, A. Strockis, J. Tsai, L. Visser, M. Wahba, S. Williams, A.H. Young, P. Ywema, S. Zisook, and E. Malievskaia

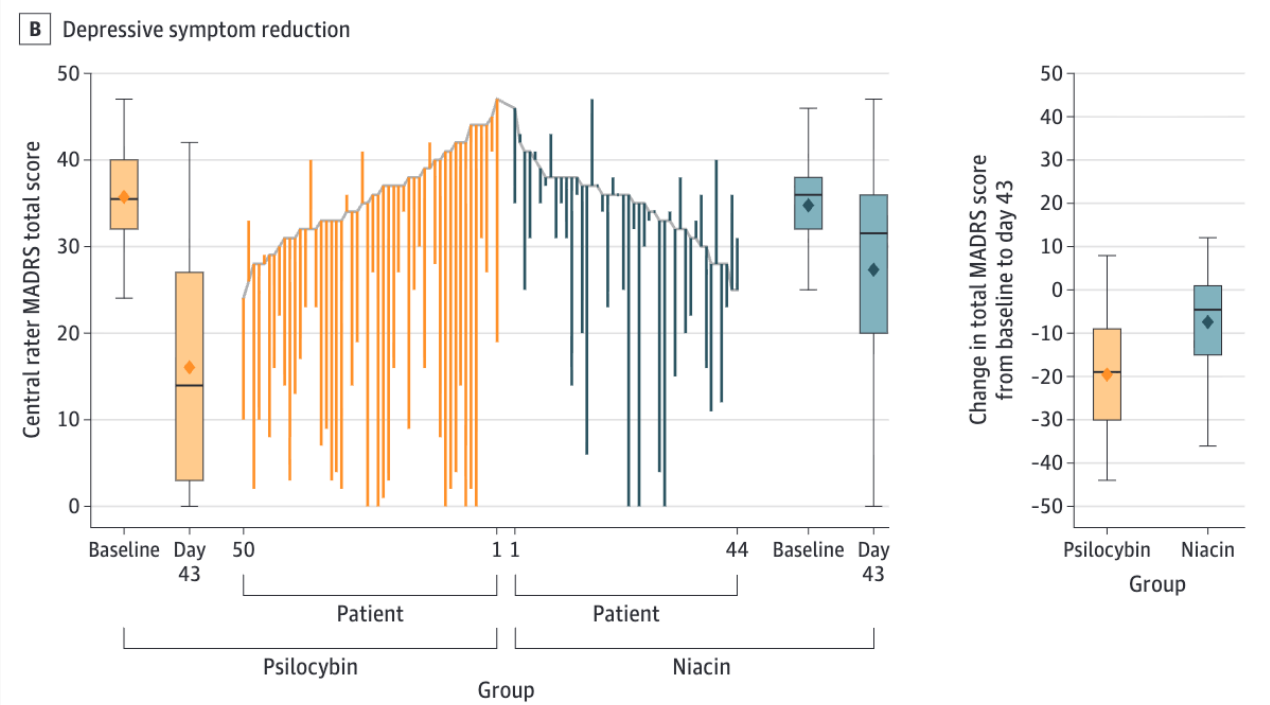
JAMA | Original Investigation

Single-Dose Psilocybin Treatment for Major Depressive Disorder
A Randomized Clinical Trial

Charles L. Raison, MD; Gerard Sanacora, MD, PhD; Joshua Woolley, MD, PhD; Keith Heinzerling, MD; Boadie W. Dunlop, MD, MS; Randall T. Brown, MD, PhD; Rishi Kakar, MD; Michael Hassman, DO; Rupal P. Trivedi, MD; Reid Robison, MD; Natalie Gukasyan, MD; Sandeep M. Nayak, MD; Xiaojue Hu, MD; Kelley C. O'Donnell, MD, PhD; Benjamin Kelmendi, MD; Jordan Sloshower, MD, MSc; Andrew D. Penn, RN, MS, NP; Ellen Bradley, MD; Daniel F. Kelly, MD; Tanja Mletzko, MA; Christopher R. Nicholas, PhD; Paul R. Hutson, PharmD; Gary Tarpley, PhD; Malynn Utzinger, MD; Kelsey Lenoch, BS; Kasia Warchol, BS; Theraysa Gapasin, MS, aMFT; Mike C. Davis, MD, PhD; Courtney Nelson-Douthit, BS; Steffanie Wilson, PhD; Carrie Brown, MA; William Linton, BS; Matthew W. Johnson, PhD; Stephen Ross, MD; Roland R. Griffiths, PhD

Psychedelics vs depression





Raison et al. JAMA. 2023;330(9):843-853. doi:10.1001/jama.2023.14530

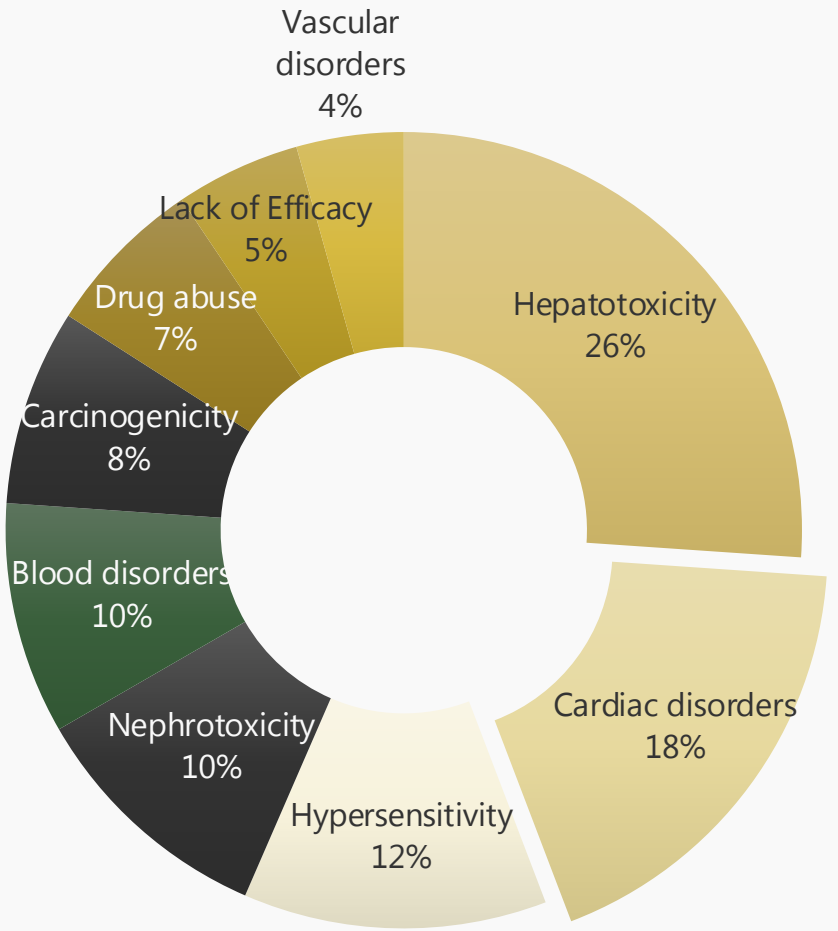
Importance of selectivity

≅33%

of clinical candidates fail
due to toxicity/adverse reactions(1)

133

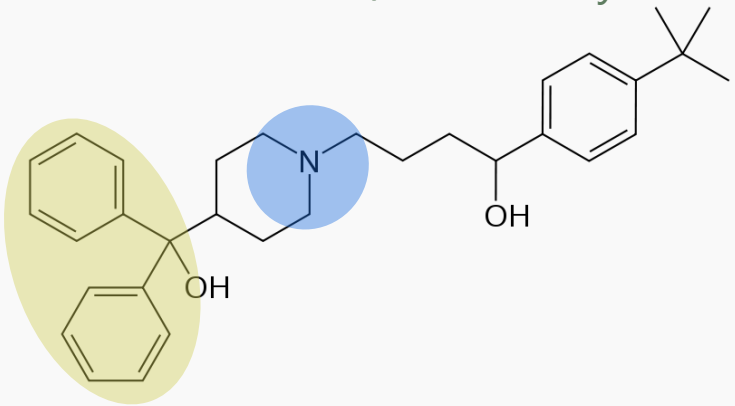
drugs withdraws between
2010-2020(2)



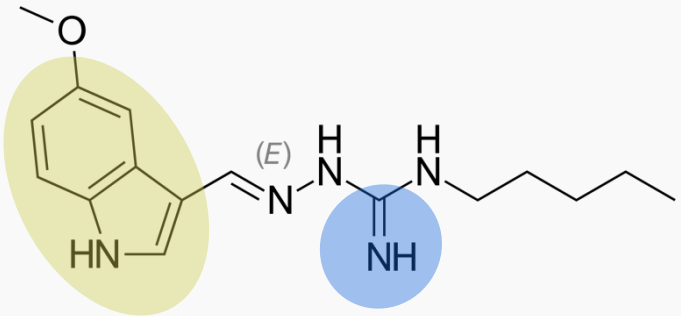
(1) D. Schuster, C. Laggner, T. Langer, Why drugs fail--a study on side effects in new chemical entities. *Curr Pharm Des* **11**, 3545-3559 (2005).
(2) N. S. Craveiro, B. S. Lopes, L. Tomás, S. F. Almeida, Drug Withdrawal Due to Safety: A Review of the Data Supporting Withdrawal Decision. *CDS*. **15**, 4–12 (2020).

Cardiac disorders

Blockage of hERG potassium channel
(cardiac arrythmia)

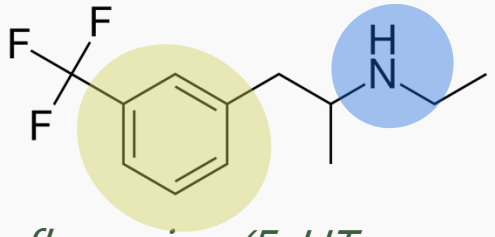


Terfenadine (histamine H1 antagonist)

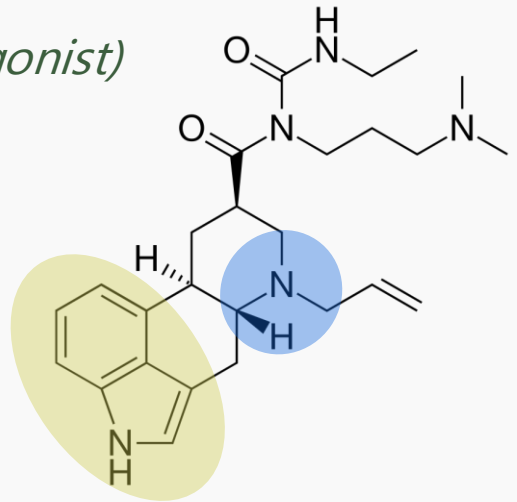


Tegaserod (5-HT$_4$ agonist)

Activation of 5-HT$_{2B}$
(valvular hearth disease)

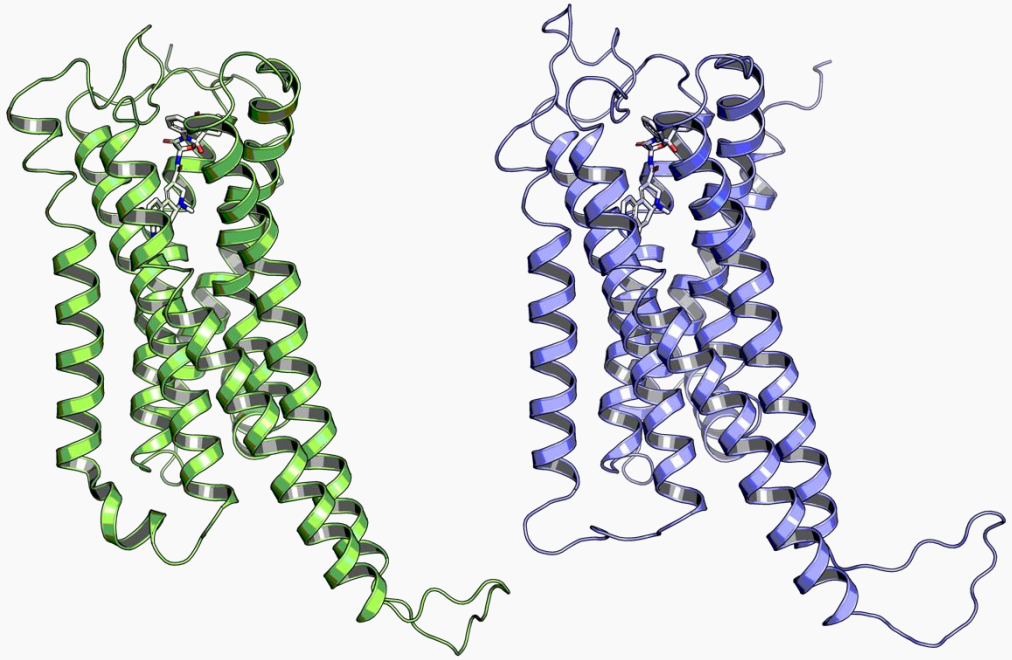


Fenfluramine (5-HT$_{2C}$ agonist)



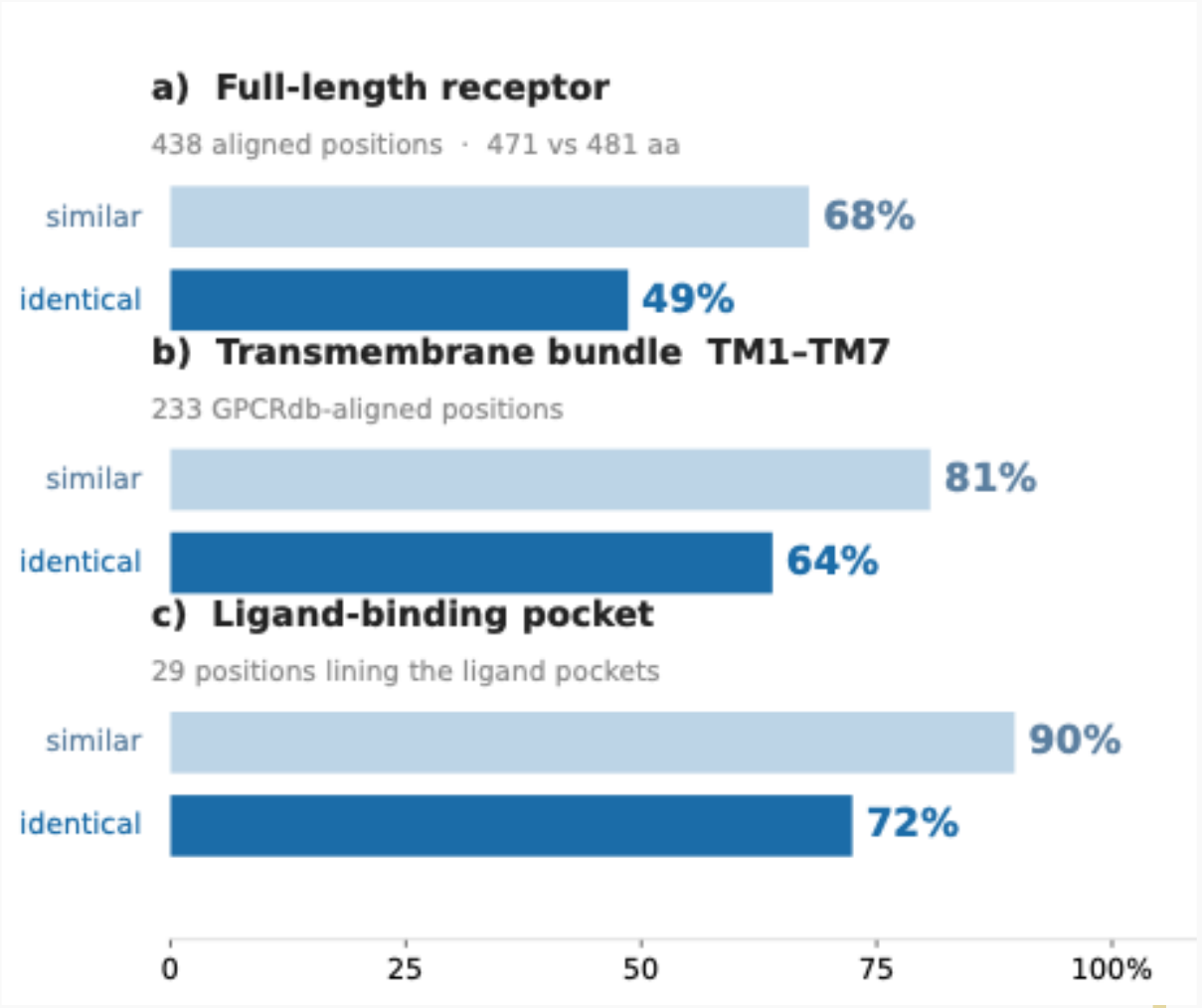
Cabergoline (dopamine D2 agonist)

5-HT_{2A} vs 5-HT_{2B}

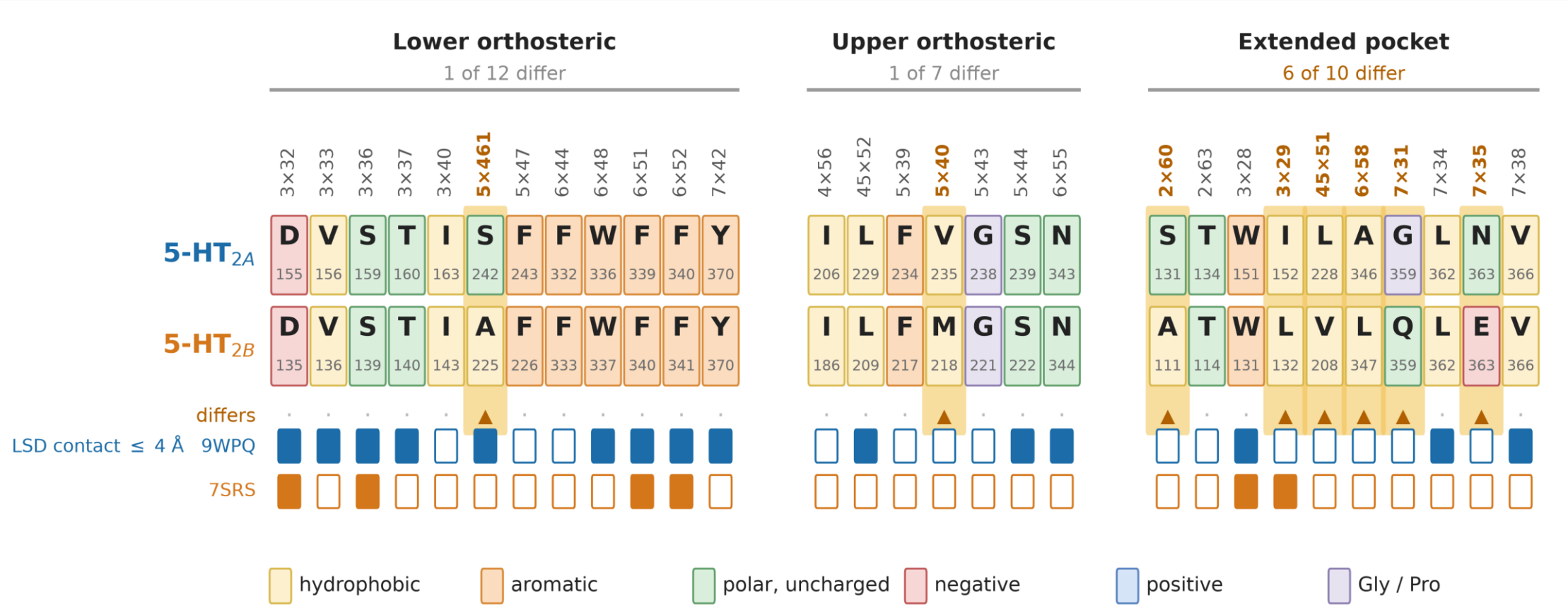


Structure of 5-HT_{2A}

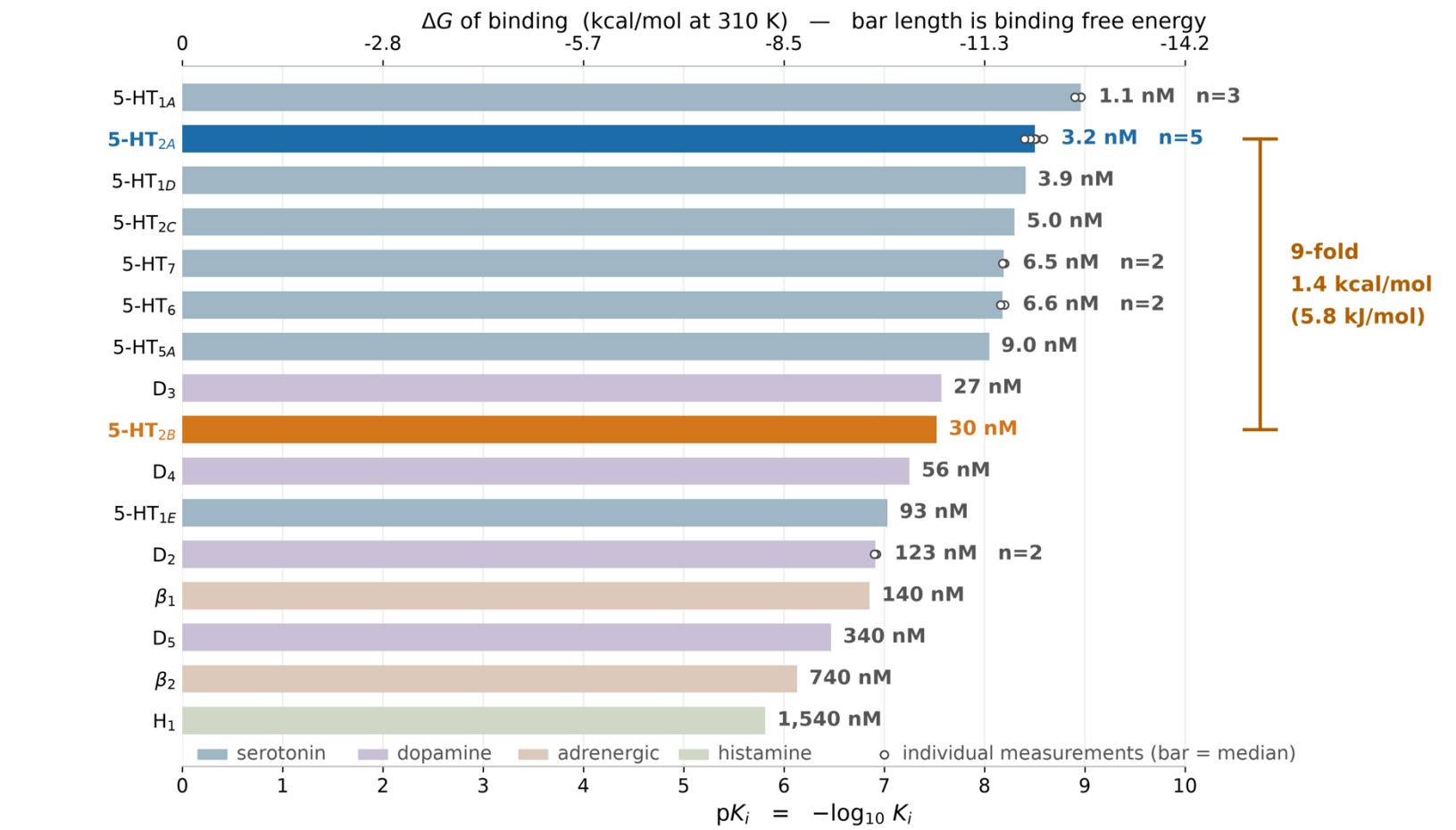
Structure of 5-HT_{2B}

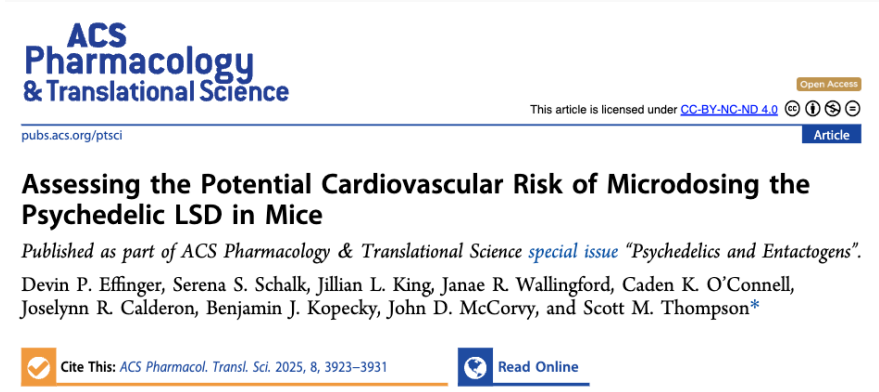


Binding pocket similarity



LSD is a (very) promiscuous drug




ACS Pharmacology & Translational Science
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pubs.acs.org/ptsci
Article

Assessing the Potential Cardiovascular Risk of Microdosing the Psychedelic LSD in Mice

Published as part of ACS Pharmacology & Translational Science special issue "Psychedelics and Entactogens".

Devin P. Effinger, Serena S. Schalk, Jillian L. King, Janae R. Wallingford, Caden K. O'Connell, Joselynn R. Calderon, Benjamin J. Kopecky, John D. McCorvy, and Scott M. Thompson*

Cite This: ACS Pharmacol. Transl. Sci. 2025, 8, 3923−3931
Read Online

Human receptors, K_i from radioligand binding, ChEMBL (LSD = CHEMBL263881; 26 values from 4 publications, 2002–2008). Values come from different laboratories, radioligands and expression systems, so small differences between receptors should not be over-read. Only these 16 aminergic receptors have human LSD K_i data in ChEMBL — α-adrenergic, muscarinic, 5-HT$_{1B}$ and 5-HT$_{1F}$ are not covered.

The Challenge

Desing a new, more 5-HT_{2A} selective LSD

- Compare the binding mode of LSD in 5-HT_{2A} and 5-HT_{2B} structures.
- Imagine changes in the LSD that can make it pick new interactions with 5-HT_{2A} and loss interaction or have clashes in the 5-HT_{2B}.
- Design and dock new compounds in the two receptor targets.
- Comparing the docking scores to estimate (roughly) binding affinity and selectivity.

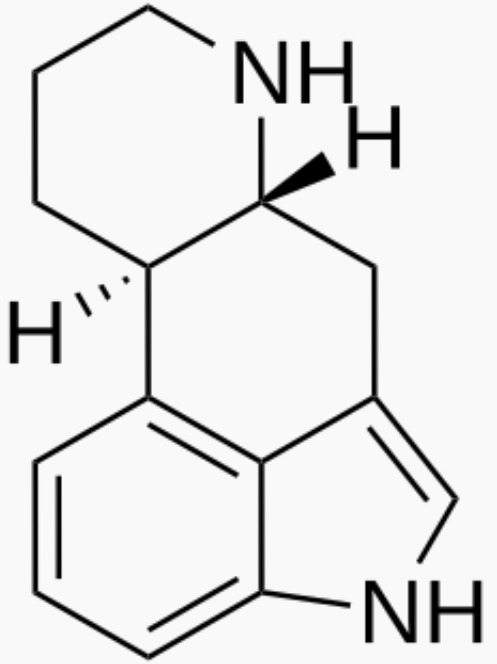
Challenge: Human vs AI chatbots

- Each group designs 3 compounds (one per person)
- Each group asks an AI chatbot (ChatGPT, Claude, Gemini, Copilot, Grok, etc.)
- We will dock and compare human vs. AI-generated designs.



⚠ Warning!

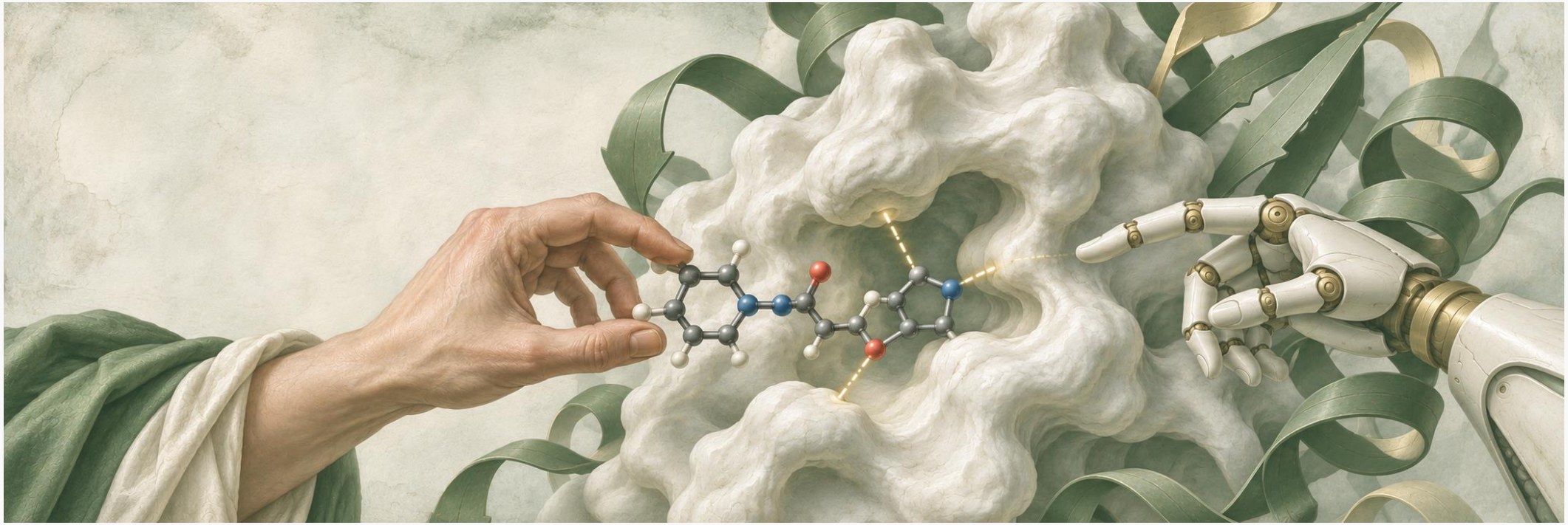
- When prompting the chatbot **do not use** the words “LSD” or “psychedelics” instead use the word for the chemical scaffold “ergolines”
- An example prompt is provided in the workshop website



Ergoline, the chemical scaffold of LSD

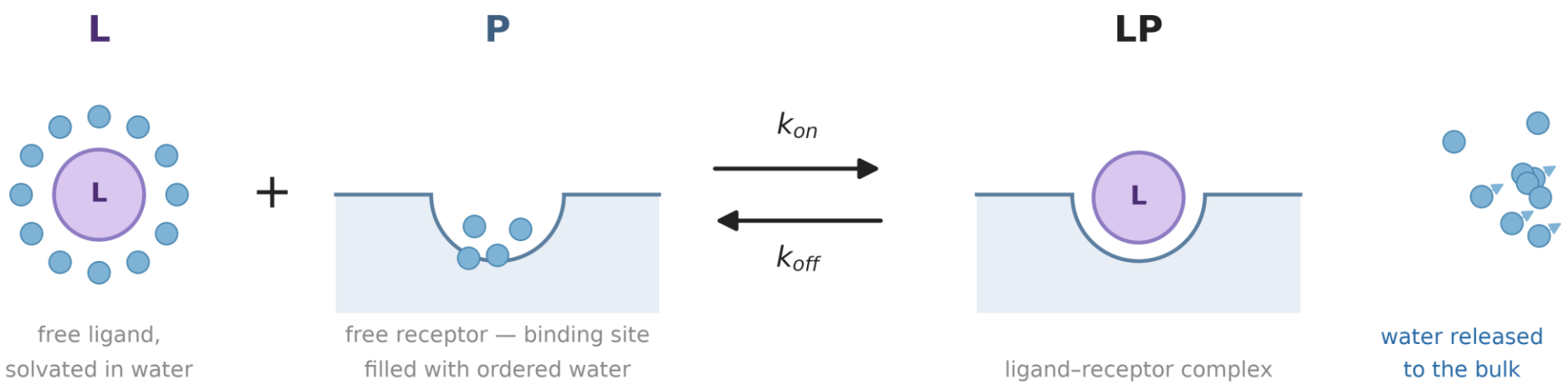
Let’s get started!

Go to the Workshop **page**
https://cpdse-edux.github.io/UCPH-PhD-AI-data-driven-drug-design/day02/index.html



Extra Info

Binding affinity and binding equilibrium



what the equilibrium defines

$$K_a = \frac{[LP]}{[L][P]} = \frac{k_{on}}{k_{off}} \qquad K_d = \frac{1}{K_a} = \frac{k_{off}}{k_{on}}$$

K_a in M^{-1} · K_d in M

what thermodynamics adds

$$\Delta G^\circ = -RT \ln K_a = +RT \ln \frac{K_d}{C^\circ}$$

$C^\circ = 1$ M standard state · $RT = 0.616$ kcal/mol at 310 K

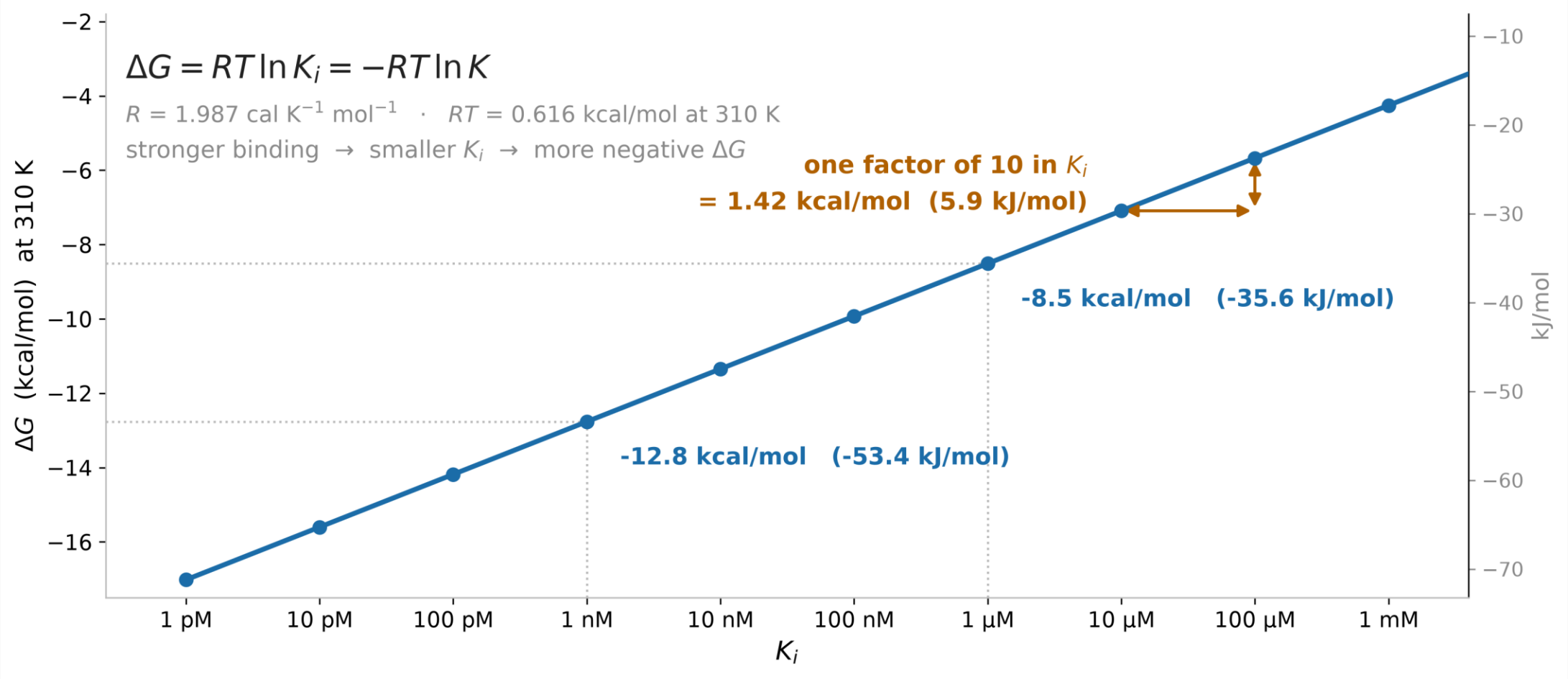
worked example

$K_d = 1$ nM $= 10^{-9}$ M ⟶ $\Delta G^\circ = 0.616 \times \ln(10^{-9}) = -12.8$ **kcal/mol** **(−53.4 kJ/mol)**

every factor of 10 in K_d = 1.42 kcal/mol (5.9 kJ/mol)

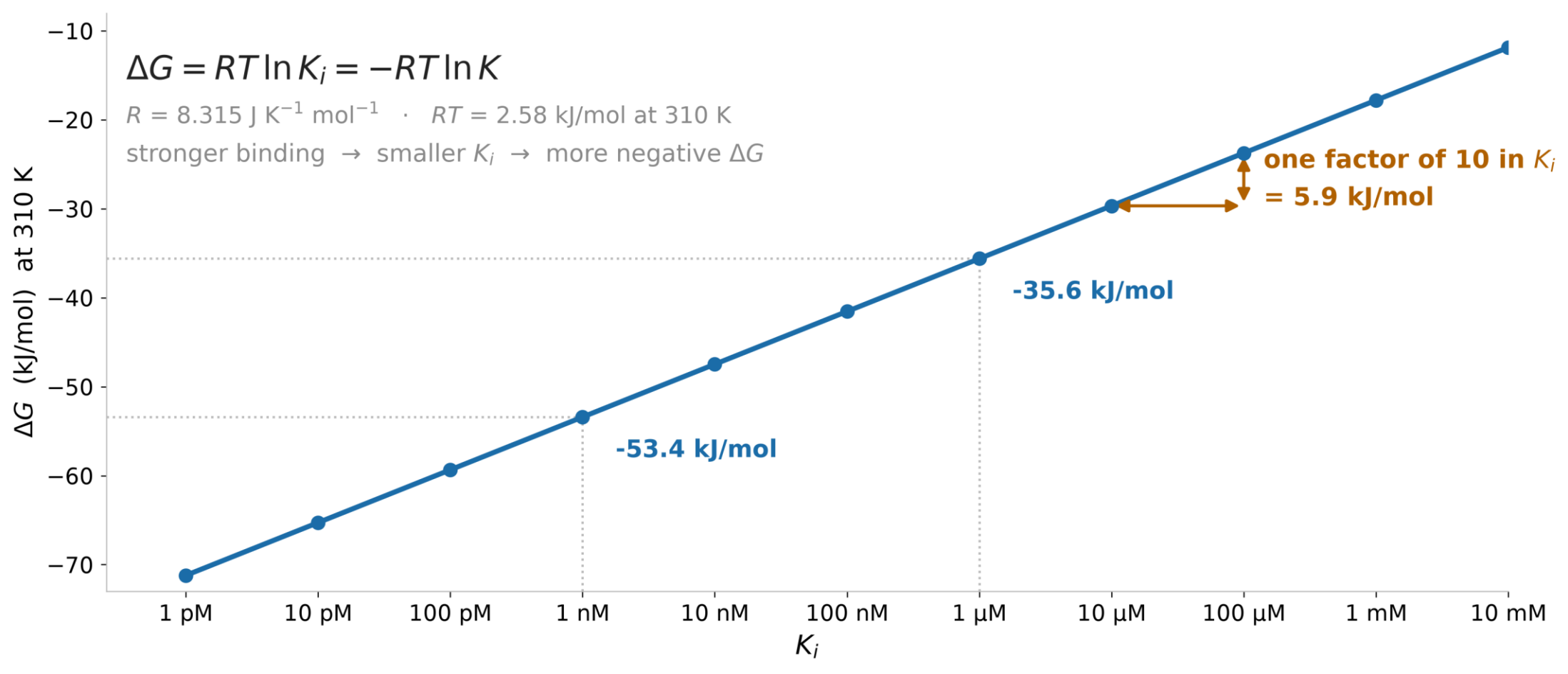
Affinity is an equilibrium, not an event: the complex constantly forms and falls apart, and K_d is the free-ligand concentration at which half the receptors are occupied. The logarithm is why affinity adds up in energy but multiplies in K_d. $K_i = K_d$ only for a competitive ligand under a suitable binding model, and a measured IC_{50} needs the Cheng–Prusoff correction, $K_i = IC_{50}/(1+[L]/K_d)$, before it can be read as an affinity.

Binding affinity and dG (in kcal/mol)



Smaller Ki = higher affinity = more negative ΔG

Binding affinity and dG (in kJ/mol)

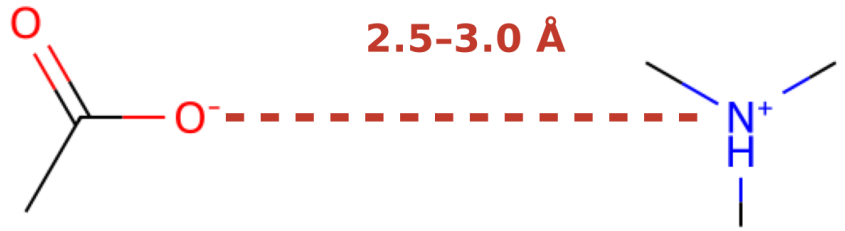


Smaller Ki = higher affinity = more negative ΔG

Interaction types: salt-bridge

2.4–4.8 kcal/mol

10–20 kJ/mol · 50- to 500-fold gain in affinity



Asp3×32 carboxylate
D155 (2A) / D135 (2B)

protonated amine
of the ligand

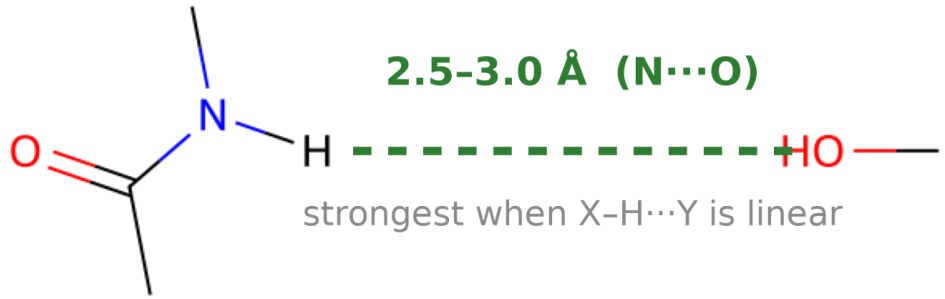
salt bridge (ion–ion + hydrogen bond)

The anchor of every aminergic ligand. In the workshop structures LSD’s protonated amine sits 3.2 Å from D3×32 in 5-HT$_{2A}$ (9WPQ) and 3.4 Å in 5-HT$_{2B}$ (7SRS) — conserved, so it buys affinity but no selectivity.

Interaction types: hydrogen bond

0.5–1.6 kcal/mol

2–6.5 kJ/mol · 2- to 13-fold gain in affinity



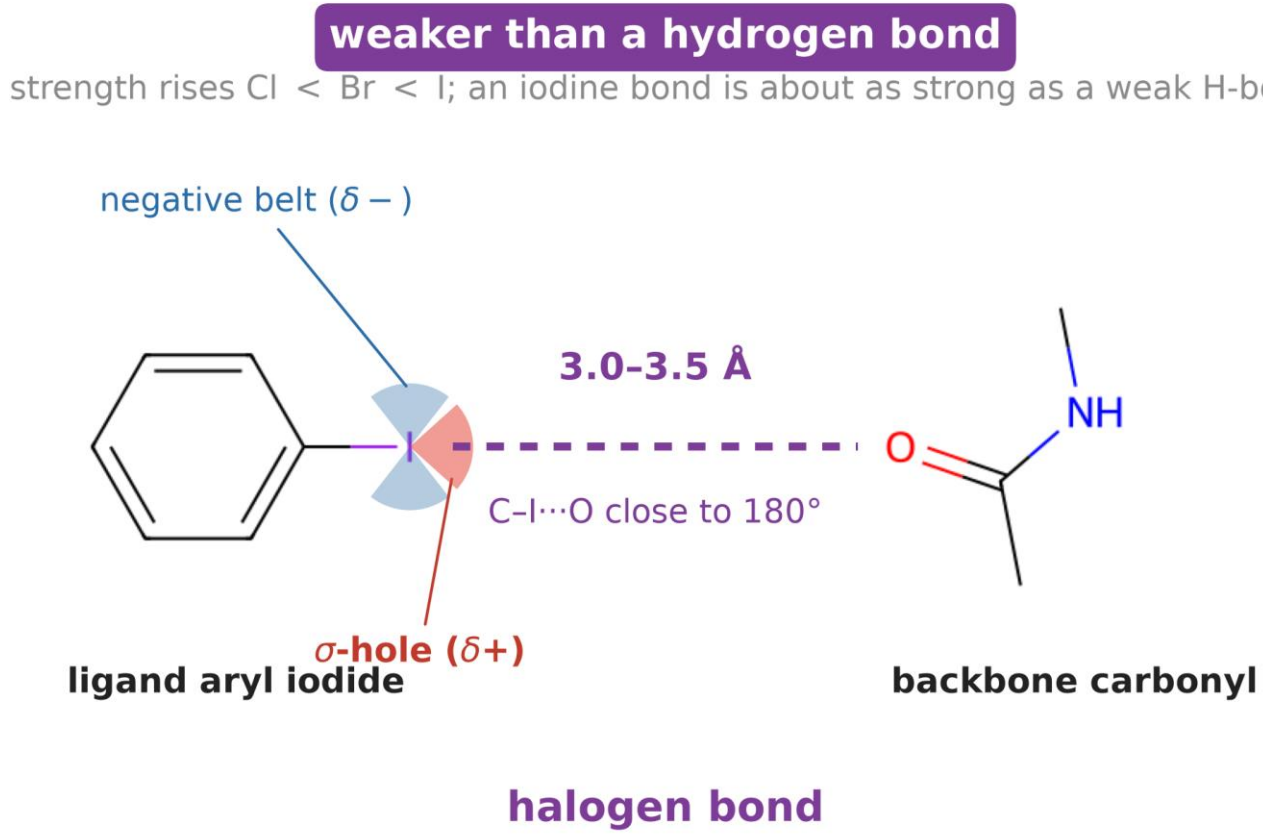
donor: amide N–H

acceptor: Ser / Thr –OH

neutral hydrogen bond

A hydrogen bond is an exchange, not a gift: both partners were already hydrogen-bonded to water. Burying a donor or acceptor without a partner costs ~1.0 kcal/mol (neutral) or ~3.8 kcal/mol (charged) — a 5- to 500-fold loss.

Interaction types: halogen bond

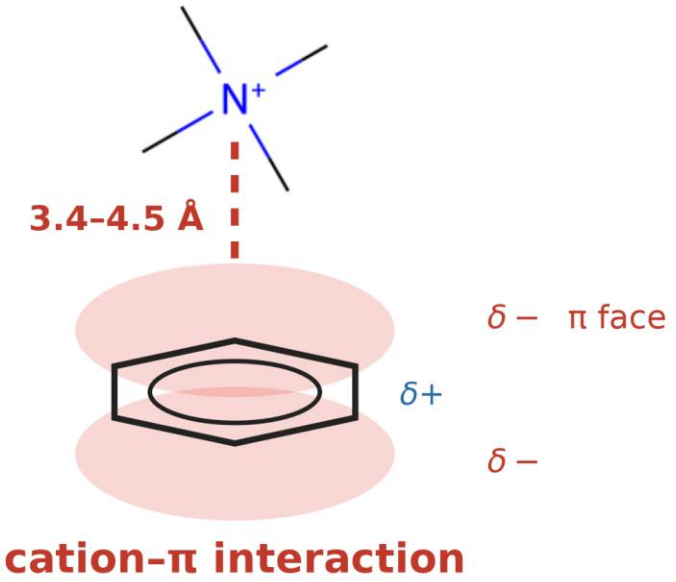


A halogen is not a sphere: three lone pairs form a negative belt around its waist, leaving an electropositive cap (σ-hole) directly behind the C–X bond, so nucleophiles approach head-on and electrophiles side-on. Iodine substitution gained 300-fold on HIV-1 reverse transcriptase (bond to the Tyr188 carbonyl) and 25-fold in a BACE series.

Interaction types: cation-pi

1.9–4.1 kcal/mol

8–17 kJ/mol · 23- to 760-fold gain



Trp6×48

Phe6×51 / 6×52

Tyr7×42

Electrostatic: the π cloud above and below an aromatic ring is electron-rich, so a cationic amine binds the ring face. Three aromatic residues line the deep 5-HT_2 pocket — W6×48, F6×51 and F6×52 — all identical in 2A and 2B. Measured inside a protein the same contact is worth less: mutating a buried Trp against Lys/Arg/His gives only 0.5–0.8 kcal/mol, because the cation must be desolvated first.

Interaction types: Pi-Pi interaction

benzene dimer: D_e = −2.5 kcal/mol calculated, −1.6 to −2.4 kcal/mol measured (7–10 kJ/mol)

the parallel-displaced and T-shaped arrangements are the two seen in protein–ligand complexes, and calculations put them essentially isoenergetic



face-to-face (sandwich)
not what two electron-rich rings do; needs an electron-poor partner

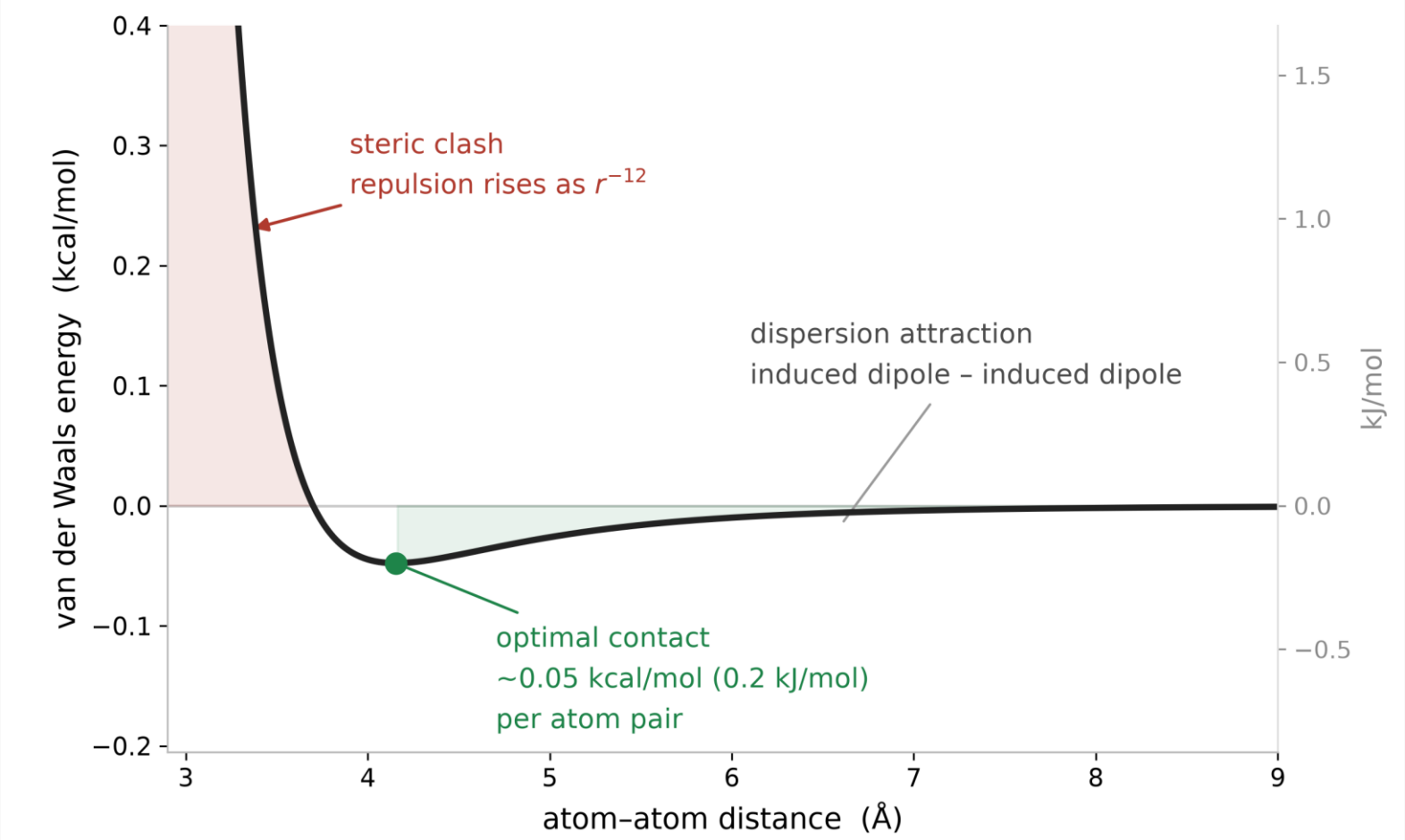
parallel-displaced
the rings slide sideways so the clouds no longer face each other

T-shaped (edge-to-face)
the δ+ C–H edge of one ring meets the δ − face of the other

amide–π
the backbone amide itself stacks on an aromatic ring

Stacking is electrostatics, not a special "π bond". Two electron-rich rings repel face-on, so they displace or turn edge-on; an electron-poor ring stacked on an electron-rich one profits from charge transfer and can sit face-to-face, and stacking between two electron-deficient rings is generally preferred over two electron-rich ones. Values and geometries from Bissantz, Kuhn & Stahl, J Med Chem 53:5061–5084 (2010).

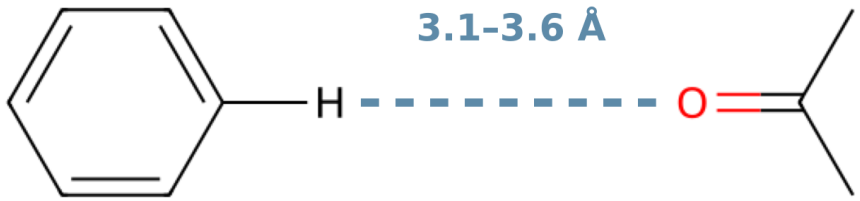
Interaction type: Van der Waals



One contact is negligible; a few hundred of them are not — 42% of all ligand–receptor contacts in serotonin-receptor structures are plain vdW / proximity contacts, and only 8% are polar.

Interaction type: weak hydrogen bond

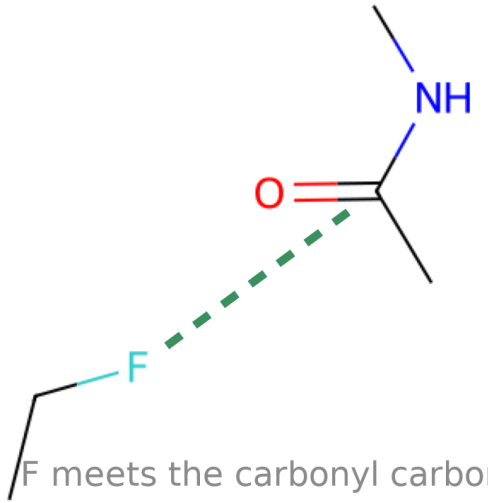
small, directional, almost free to form



an aromatic C–H is the donor

weak hydrogen bond C–H···O

ΔΔ*G* = −0.2 to −0.3 kcal/mol (0.8–1.3 kJ/mol)



F meets the carbonyl carbon at ~90°

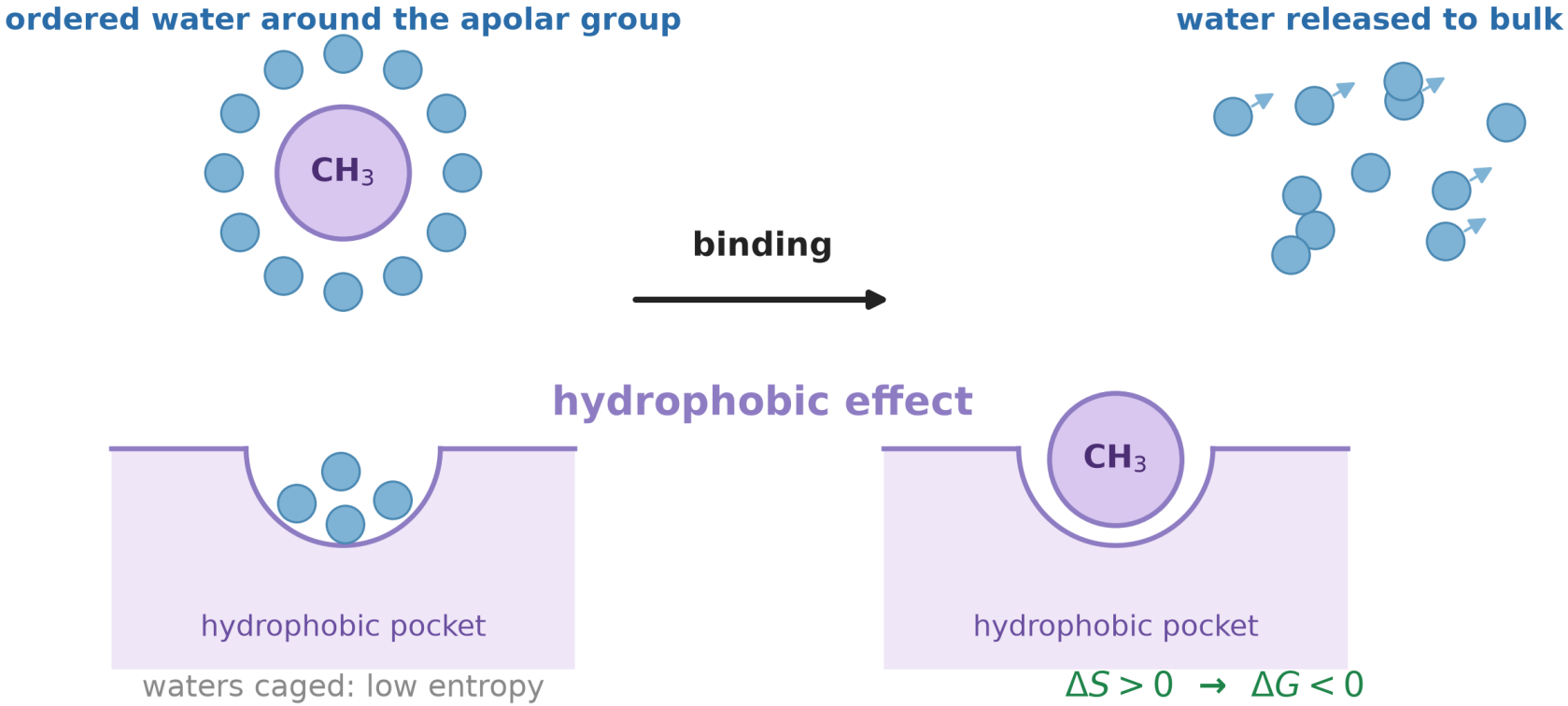
orthogonal multipolar C–F···C=O

Both are weak, but they cost almost nothing to make: unlike a classical hydrogen bond, a C–H donor or a C–F acceptor is barely solvated in water, so little has to be stripped away first. They are never repulsive and are directional enough to steer a pose.

Hydrophobic effect

0.024–0.057 kcal/mol per Å^2 buried

0.1–0.24 kJ/mol per Å^2 · one CH_3 (~25 Å^2): 0.6–1.4 kcal/mol, a 3- to 10-fold gain

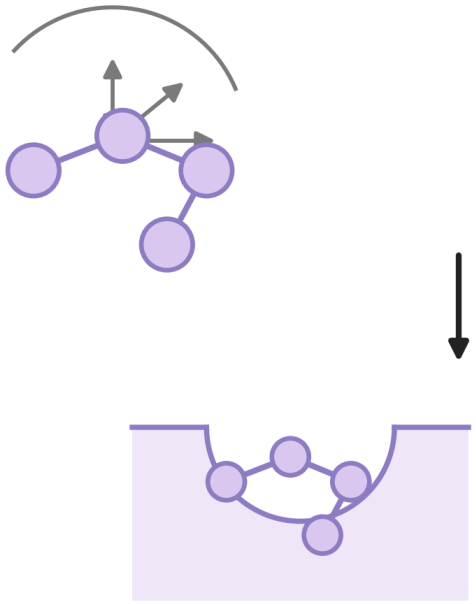


Entropy-driven: burying apolar surface frees the water that had to cage it. Not a bond — it scales with buried area, which is why overall shape complementarity outweighs any single polar contact.

Entropy cost

2.9–10.8 kcal/mol COST
12–45 kJ/mol

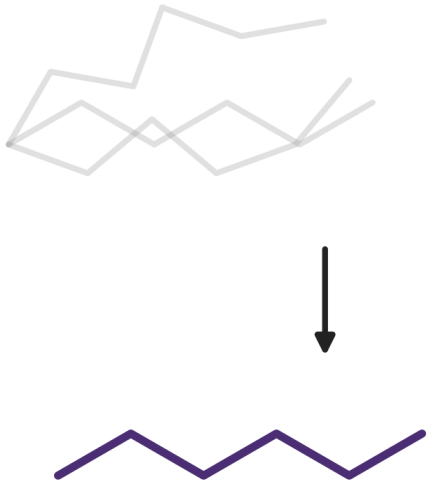
free in water:
3 translations + 3 rotations



bound: motion frozen

0.2–1.4 kcal/mol per frozen bond
1–6 kJ/mol

many conformers
in water



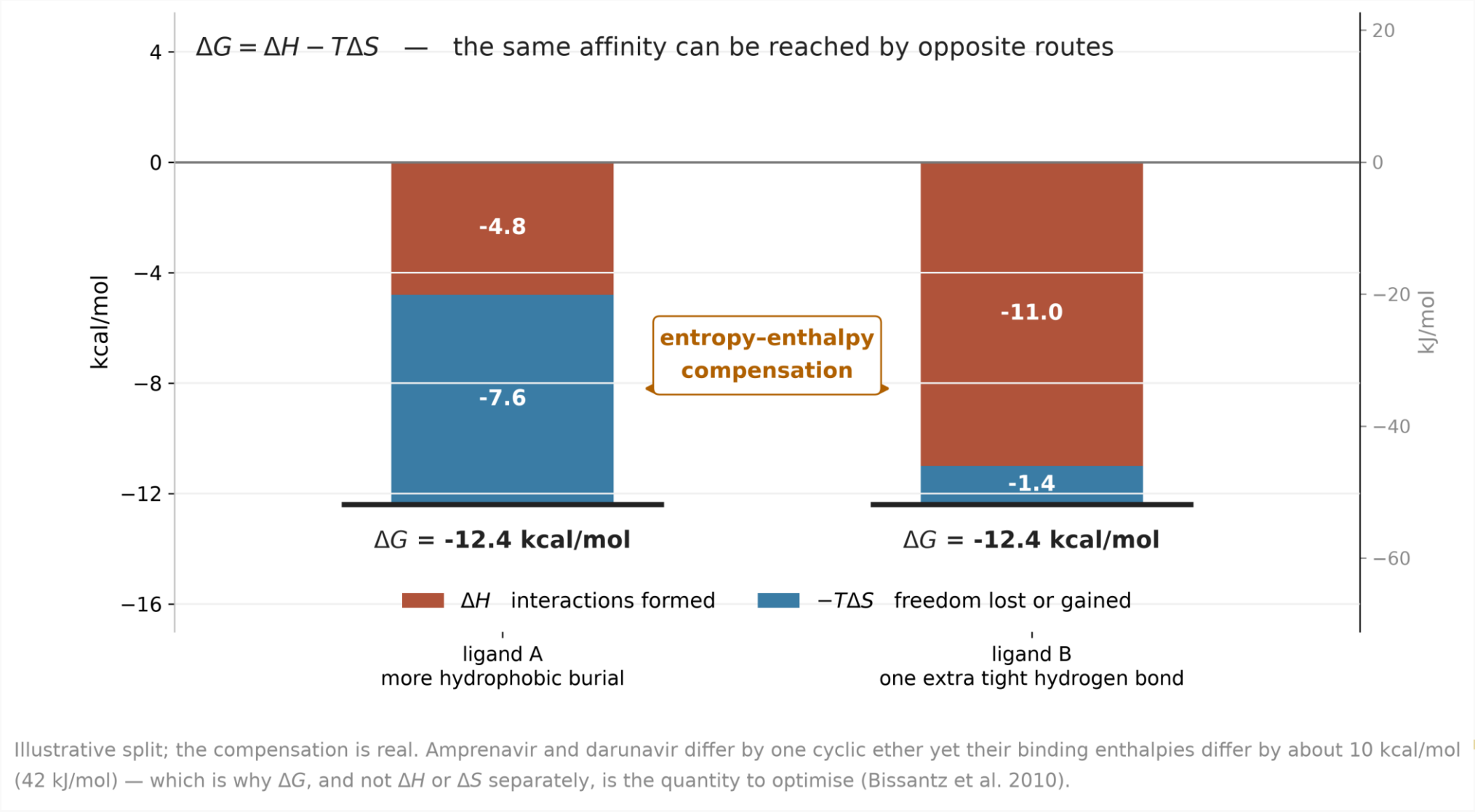
one bioactive conformation

plus strain if the bound conformer is not the preferred one
(usually < 3.1 kcal/mol; palmitic acid +2.5 kcal/mol)

These two terms always oppose binding and cannot be designed away — they must be paid back by the favourable terms.

Rigid ligands pay less: fewer rotatable bonds frozen, no strain penalty.

Enthalpy and Entropy



Illustrative split; the compensation is real. Amprenavir and darunavir differ by one cyclic ether yet their binding enthalpies differ by about 10 kcal/mol (42 kJ/mol) — which is why ΔG, and not ΔH or ΔS separately, is the quantity to optimise (Bissantz et al. 2010).

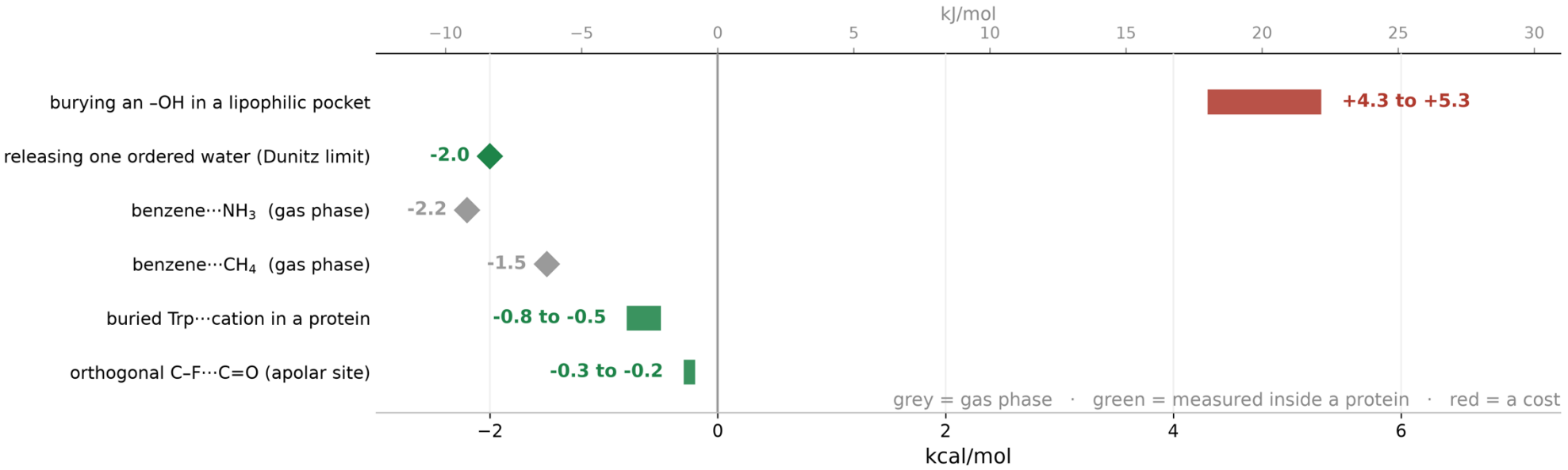
Desolvation

$$\Delta G_{net} = \Delta G_{interaction} - \Delta G_{desolvation}$$

intrinsic attraction between the two groups — **stripping the water off both partners** = **net ΔG**

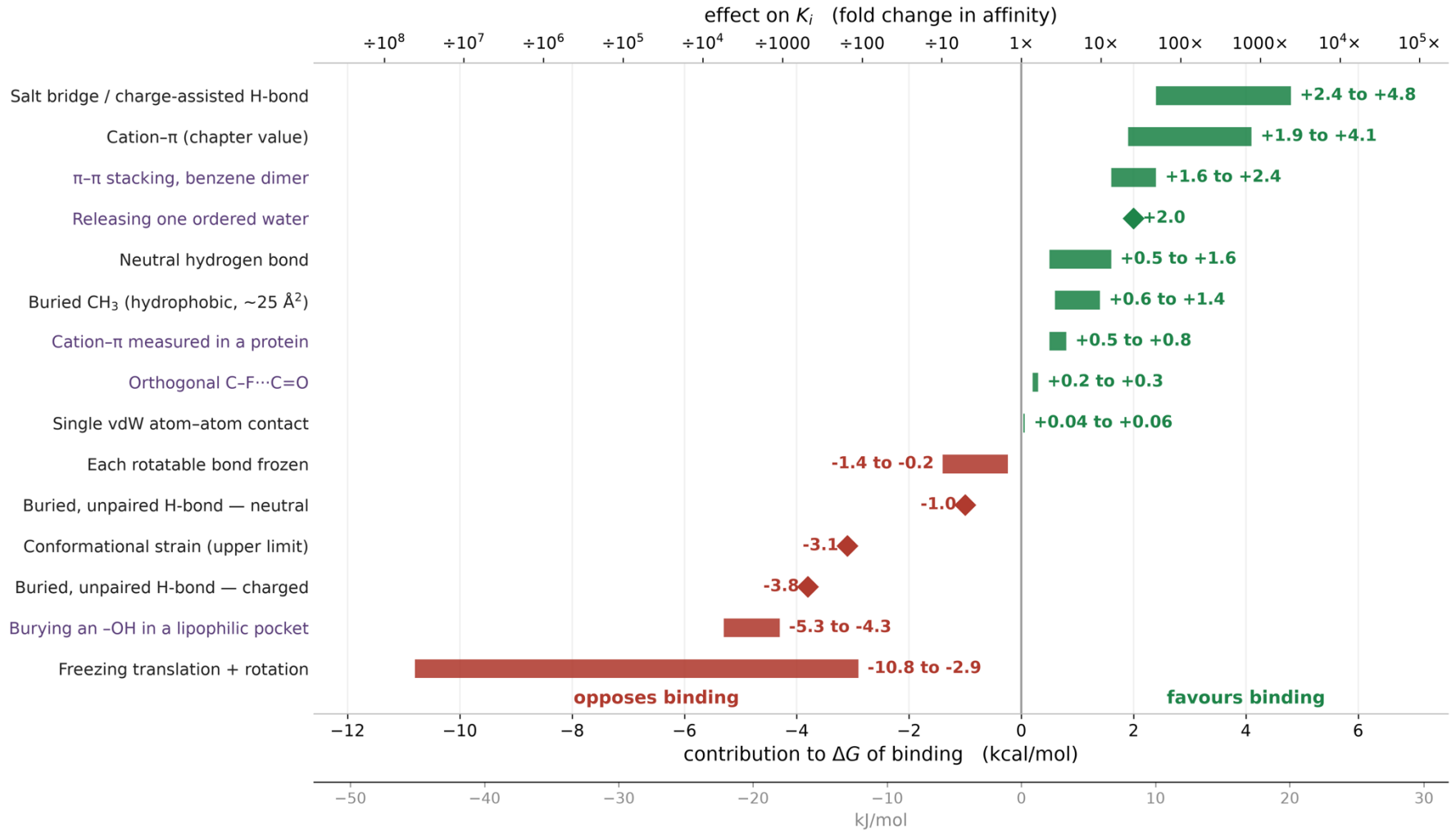
Two large numbers nearly cancel; what survives is small — and that is what K_i reports.

A “stronger” interaction is therefore not automatically a better one: the stronger partner was also the better solvated one.



Values from Bissantz, Kuhn & Stahl, J Med Chem 53:5061–5084 (2010); water-release limit from Dunitz. 1 kcal/mol = 4.184 kJ/mol.

Contribution to binding



Black labels: Liljefors, Textbook of Drug Design and Discovery Ch. 1. Purple labels: Bissantz, Kuhn & Stahl, J Med Chem 53:5061 (2010). Fold change = $10^{\Delta\Delta G/1.42}$ at 310 K. Note the two cation–π rows: the textbook range is an intrinsic estimate, the lower one is what mutating a buried Trp actually delivers once desolvation is paid.

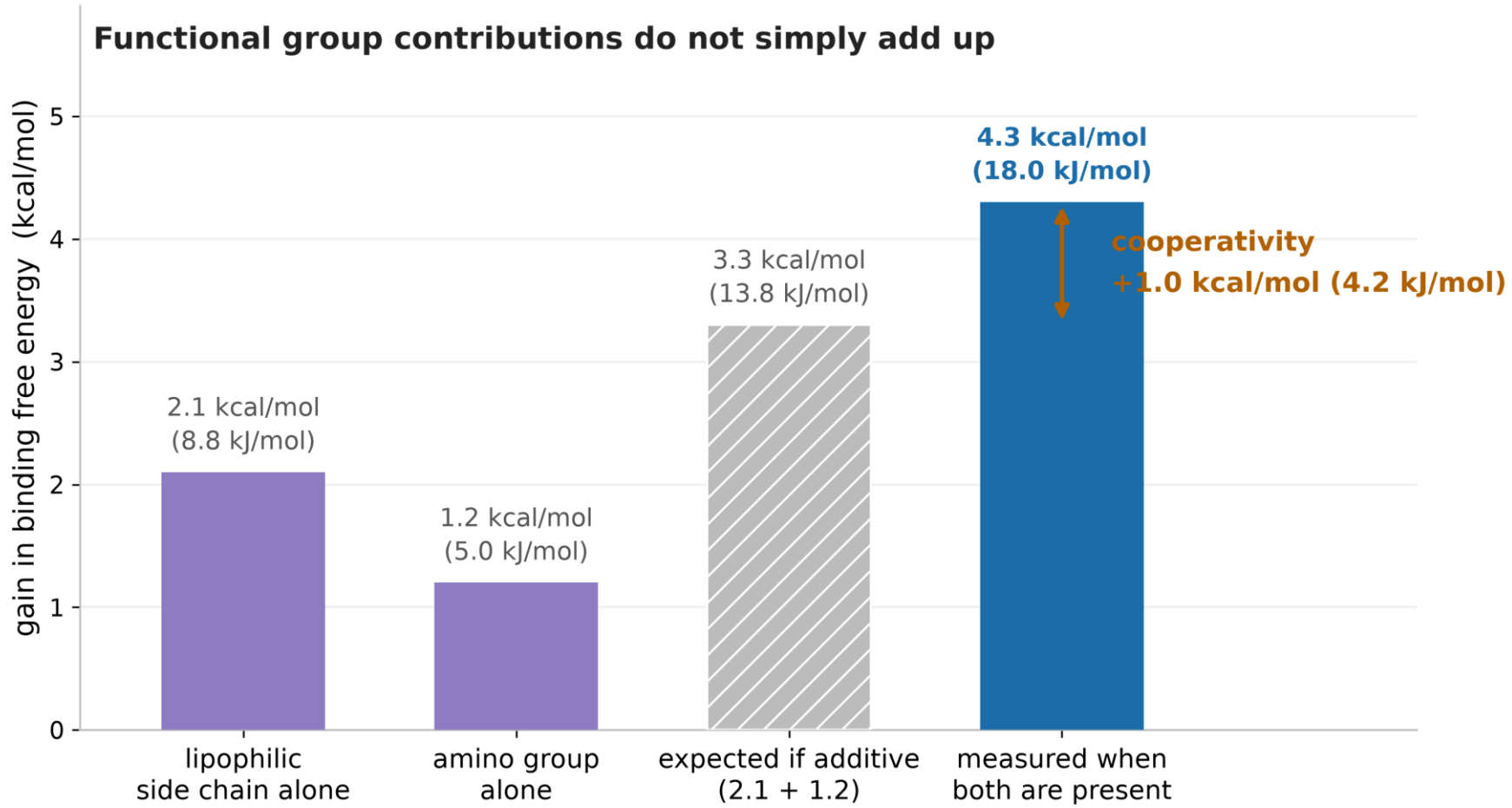
Contribution to binding

Interaction or term	Typical geometry	ΔG (kcal/mol)	ΔG (kJ/mol)	Effect on K_i	Desolvation	Src
Salt bridge / charge-assisted H-bond	2.5–3.0 Å, directional	**+2.4 to +4.8**	+10 to +20	50–500×	very high	L
Neutral hydrogen bond	2.5–3.0 Å, X–H···Y ≈ 180°	**+0.5 to +1.6**	+2 to +6.5	2–13×	high	L
Cation–π (textbook estimate)	3.4–4.5 Å over the ring face	**+1.9 to +4.1**	+8 to +17	23–760×	moderate	L
Cation–π (buried Trp, by mutation)	—	**+0.5 to +0.8**	+2 to +3	3–4×	moderate	B
π–π parallel-displaced	3.3–3.8 Å, offset 1.2–1.8 Å	**+1.6 to +2.4**	+7 to +10	—	moderate	B
π–π T-shaped (edge-to-face)	70–90° between planes	**same as above**	same	—	moderate	B
π–π face-to-face (sandwich)	~3.5 Å, rings eclipsed	**repulsive (electron-rich pair)**	—	—	moderate	B
Amide–π	3.2–3.7 Å plane-to-plane	**like an H-bond**	like an H-bond	—	low	B
Halogen bond C–X···O	3.0–3.5 Å, ≈180°, Cl < Br < I	**weaker than an H-bond**	—	up to 300×	low	B
Weak hydrogen bond C–H···O	3.1–3.6 Å, mildly directional	**small, never repulsive**	—	—	very low	B
Orthogonal multipolar C–F···C=O	F to carbonyl C at ≈90°	**+0.2 to +0.3**	+0.8 to +1.3	—	very low	B
Hydrophobic burial	shape complementarity	**+0.024 to +0.057 / Å^2**	+0.1 to +0.24 / Å^2	CH_3: 3–10×	none	L / B
van der Waals / dispersion	3.6–4.2 Å C···C	**+0.05 per atom pair**	+0.2 per pair	only in sum	none	L
Releasing one ordered water	—	**up to +2.0**	up to +8.4	—	a gain	B
Buried unpaired polar group	donor or acceptor, no partner	**−1.0 / −3.8**	−4 / −16	5–500× loss	the penalty	L
Buried –OH in a lipophilic pocket	—	**−4.3 to −5.3**	−18 to −22	—	the penalty	B
Each rotatable bond frozen	—	**−0.2 to −1.4**	−1 to −6	—	—	L
Freezing translation + rotation	whole molecule, one-off	**−2.9 to −10.8**	−12 to −45	—	—	L

Src: L = Liljefors, Textbook of Drug Design and Discovery (310 K) · B = Bissantz, Kuhn & Stahl, J Med Chem 53:5061 (2010). 1 kcal/mol = 4.184 kJ/mol.

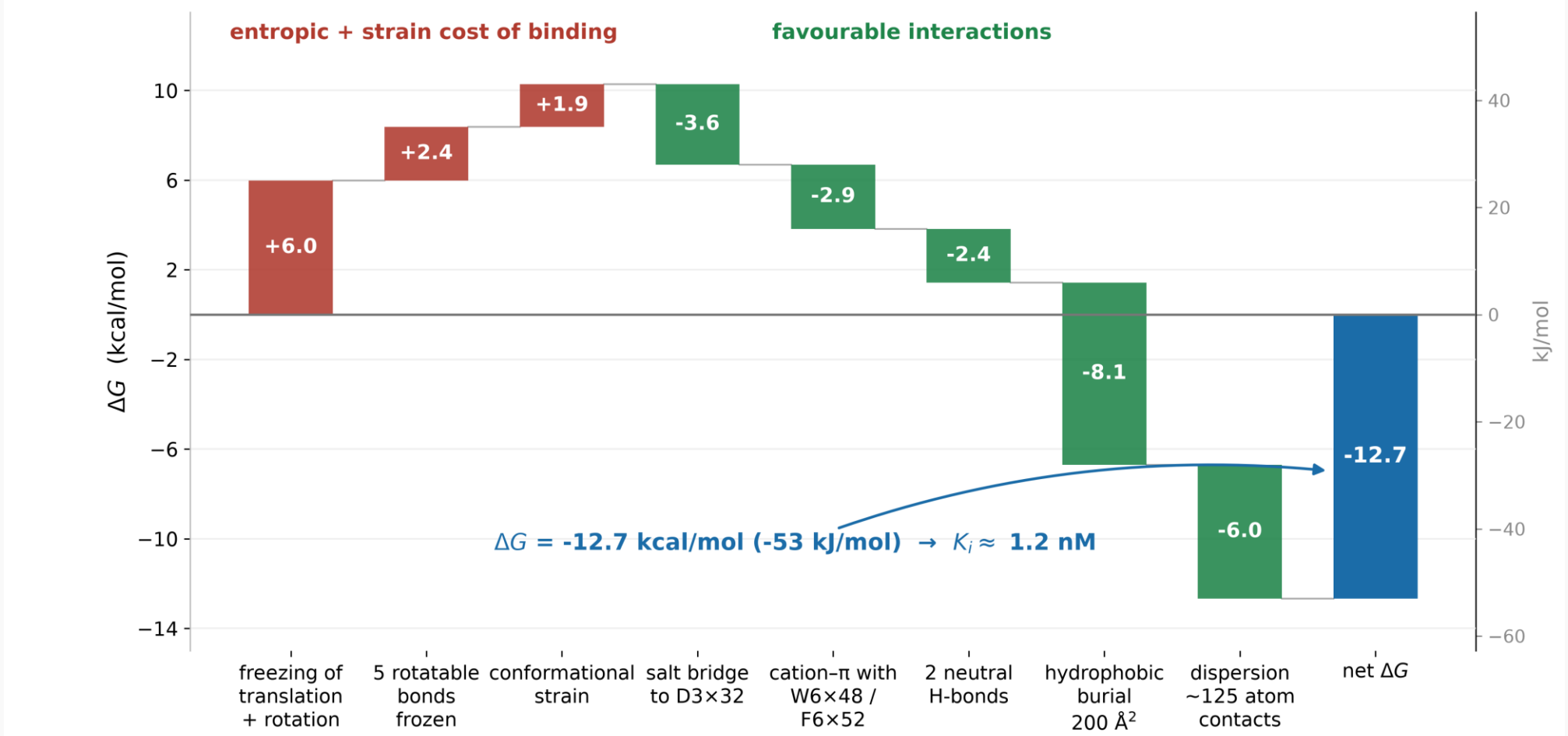
Positive = favours binding. Every 1.42 kcal/mol (5.9 kJ/mol) is one order of magnitude in K_i. These are typical values, not constants — the same group is worth different amounts in different pockets.

However…



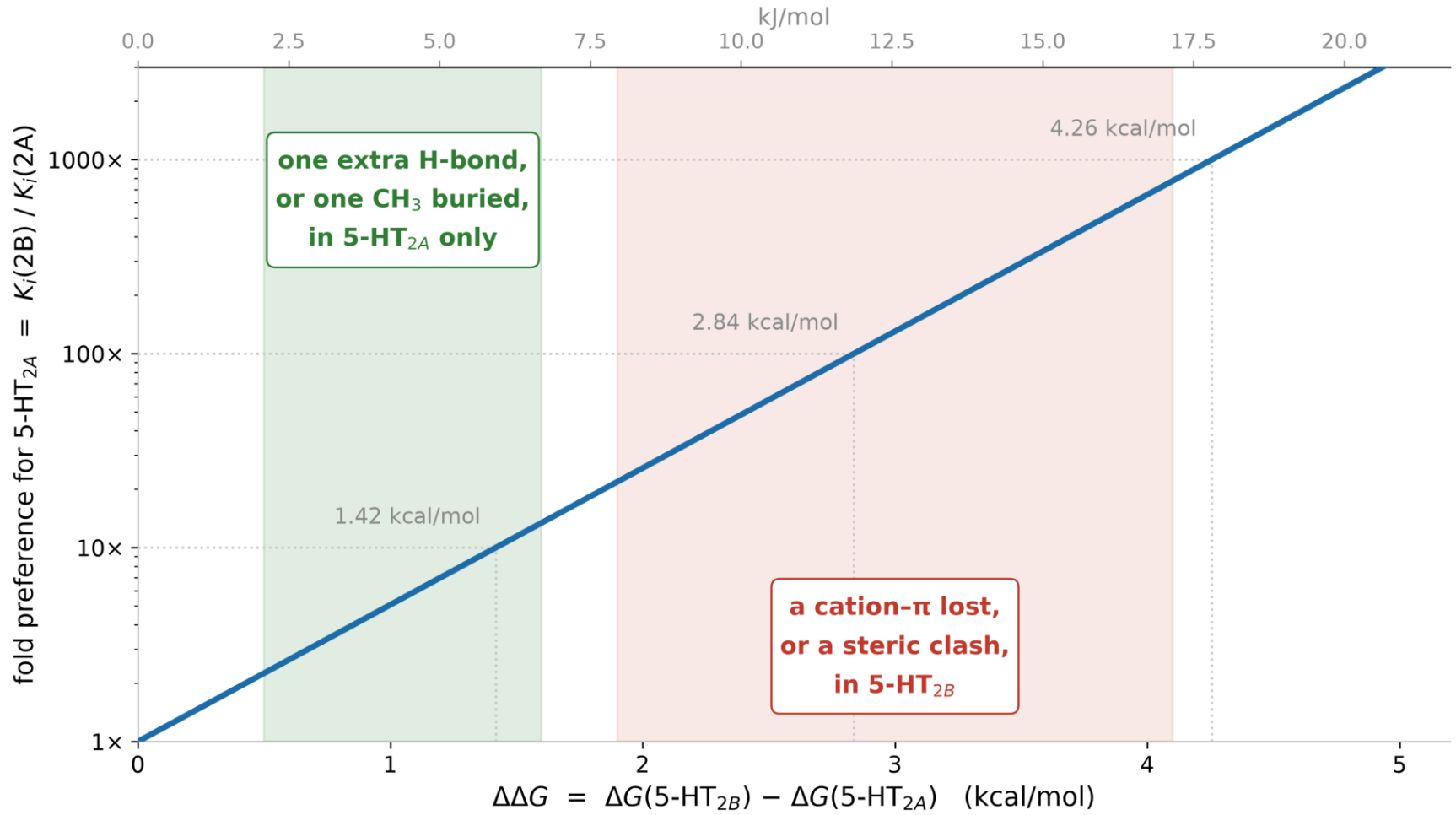
Thrombin inhibitor series analysed by the Klebe group, as presented in Bissantz et al. 2010 (Figure 1). Forming one interaction tightens the complex, so the next one is worth more than it would be on its own — an energy bookkeeping that assigns a fixed value per group will be wrong.

Binding energy budget: LSD example



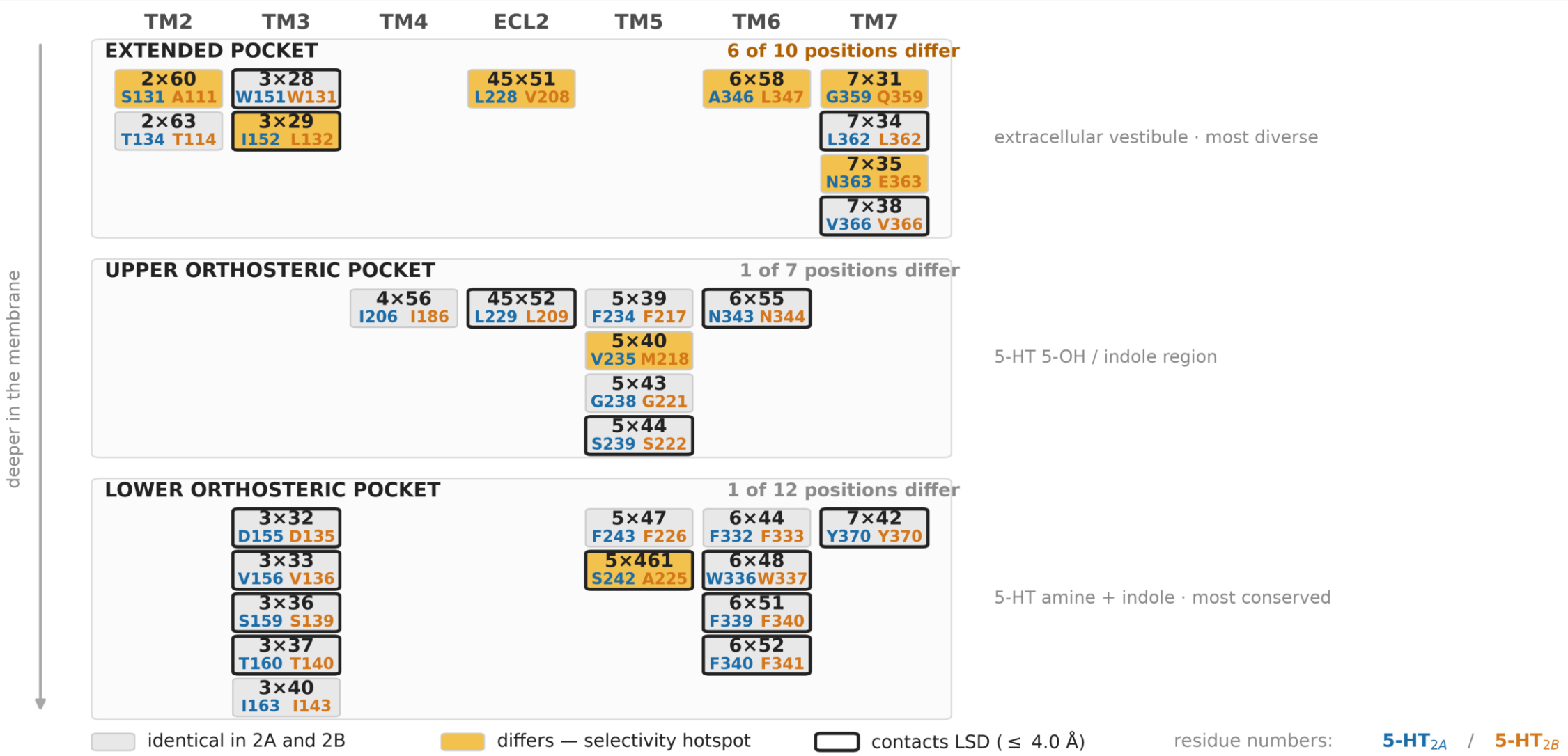
The favourable terms deliver 23 kcal/mol (96 kJ/mol); 10.3 kcal/mol (43 kJ/mol) is spent on freezing motion and strain. Illustrative budget built from the ranges in Liljefors, Textbook of Drug Design and Discovery — the message is the size of the entropic bill, not the exact split.

Improving selectivity LSD example

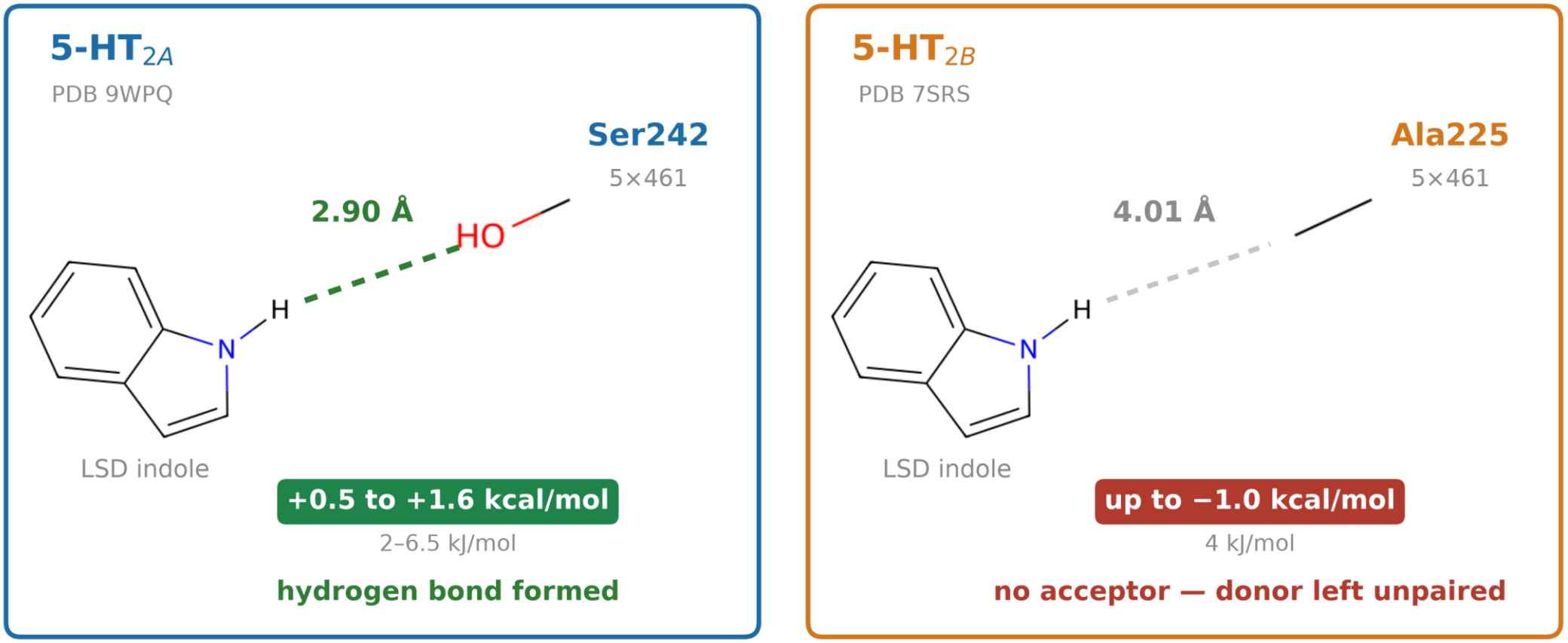


Selectivity obeys the same arithmetic as affinity: every 1.42 kcal/mol (5.9 kJ/mol) of differential binding free energy is one order of magnitude of subtype preference. Modest, well-placed differences suffice — and repulsion in the off-target counts as much as attraction in the target.

Extra: Binding pocket similarity



LSD: the S242A



ΔΔ*G* ≈ 1.4–2.4 kcal/mol (6–10 kJ/mol) → roughly 10- to 50-fold preference for 5-HT$_{2A}$

S5×461 is the only binding-site residue unique to 5-HT$_{2A}$ within the 5-HT$_2$ subfamily; the penalty in 2B applies only if no water takes the serine's place